Statistically Modelling Qualia Syntax Through Fourier Space: A Three-role Probabilistic Framework Where Ambiguity Becomes Structure

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Abstract: This article investigates whether composers' deployment of harmonic qualities—including whole-tone quality and diatonicity—exhibits systematic patterns that could constitute a qualia syntax. Whilst tonal functional harmony possesses well-established syntactical rules, the fundamental principles underlying musical syntax in 20th-century compositional practices remain largely unexplored. Building upon the discrete Fourier transform of pitch-class sets and its geometric interpretation through the Fourier Qualia Space (FQS), we advance three key contributions: (1) the definition of an ambiguity region in the FQS, (2) a computational methodology for extracting qualia progressions from musical works, and (3) a four-tiered nested analysis examining these progressions across multiple analytical scales. Our findings demonstrate an evolution towards language-like distributions in qualia deployment over the centuries, with Impressionistic compositional practices serving as a notable example of this trend. Moreover, through detailed examination of Debussy's works, we identify a three-role model wherein harmonic qualities serve distinct functions: qualia anchor, mediating transitional, and colouristic contrast roles. These results reveal organised patterns in qualia treatment and establish a statistically supported framework for understanding the key relationships between harmonic qualities in Debussy's music.

Keywords: Musical Syntax, Discrete Fourier Transform, Harmonic Qualities, Impressionism, Language, Zipf's Law, Statistics.

1. Introduction

Intuitively, when individuals communicate through speech or writing, they employ phrasal construction rules that enable them to express their ideas effectively. Without conscious awareness, they commonly structure sentences beginning with subjects, followed by predicates or direct objects, among other syntactic functions (Chomsky, 2015, pp. 73–80; Jackendoff, 2004, pp. 9—10). This set of rules forms the foundation of linguistic syntax, which seeks to explain how sequential words construct complex ideas and actions. Although musical syntax differs fundamentally from linguistic syntax, music similarly encompasses rules and functions that provide composers with the structural logic to create intelligible musical phrases—complete with directionality, growing expectation, resolution, beginning, and conclusion (Lerdahl, 2013; Rohrmeier & Pearce, 2018). The parallel period from the Classical era exemplifies such musical syntactic constructs. In these structures, composers typically craft two musical phrases in a question-and-answer format, traditionally termed antecedent and consequent, both initiating with tonic prolongation (Caplin, 2001, pp. 49–58; Forte & Gilbert, 1982, pp. 142-158; Schenker & Oster, 2001, pp. 68-70). The composer then develops the musical phrase
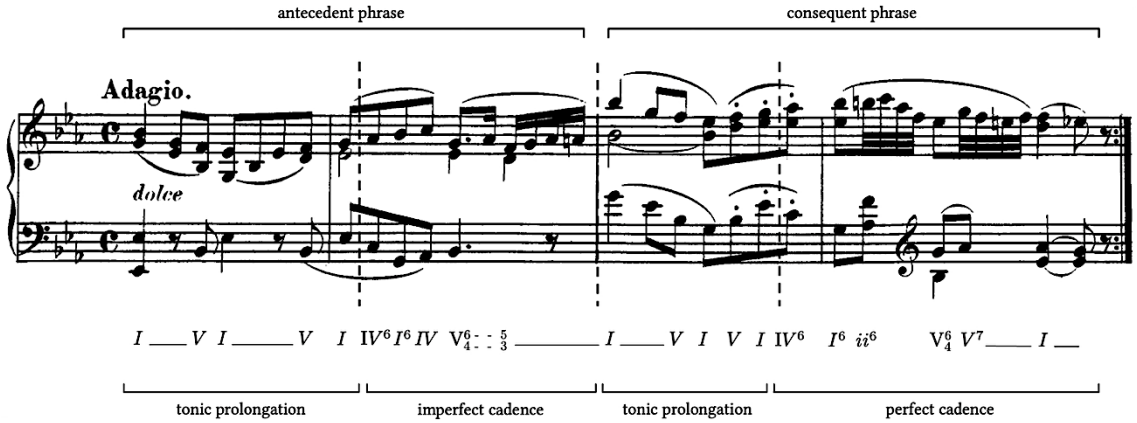


Figure 1. The formal and syntactical structures of a parallel period in the opening of Mozart's Piano sonata KV. 570, 2nd movement (mm. 1—4). The symmetrical shape of the period stems from two identical phrases comprised of tonic prolongation and two cadential movements—at first, imperfect, for an unresolved and inquiring phrase, and lastly, perfect, for an answer.

by incorporating a half cadence in the first phrase and a perfect cadence in the second.[1] For a comprehensive diagram of the parallel period, Figure 1 demonstrates its structural division and intrinsic chord progression.

However, musical syntax remains insufficiently systematised, particularly regarding 20th-century aesthetic movements (Hasty, 1984; Lerdahl, 2001, p. 344). This systematisation challenge becomes even more pronounced when examining harmonic qualities such as diatonicity or octatonicity, where the existence of governing syntax remains largely unexplored.[2] Specifically, the question of whether 20th-century composers' deployment of harmonic qualia—a term we employ, following Harding (2021), to denote harmonic qualities such as diatonicity or whole-tone quality, rather than the subjective consciousness experiences of philosophical discourse (Pereira et al., 2024; Pereira et al., 2025)—follows systematic rules that could constitute a qualia syntax remains unanswered (Mailman, 2015). The significance of this question varies across compositional practices, as harmonic qualia hold markedly different structural importance in, for instance, tonal functional harmony versus Bartók's compositions or even atonal works.[3] Consequently, computational approaches—both general computing and computational musicology specifically—have recently focused on musical syntax, with the former modelling existing knowledge and the latter investigating novel syntactic organisational structures across diverse musical styles (Pearce & Rohrmeier, 2018).

The discrete Fourier transform (DFT) of pitch-class sets (pcsets) provides a particularly relevant mathematical framework to our work, as literature has established clear associations between its six non-trivial coefficients and six distinct harmonic qualities: chromaticity, dyadicity, triadicity, octatonicity, diatonicity, and whole-

[1] As the first degree is the point of greatest harmonic stability, half cadences, as the name suggests, keep the listener in suspense because they do not produce a sense of full and truly conclusive phrase ending, almost as if the phrase were a question requiring an answer (Caplin, 2001, pp. 49-58).

[2] One work that suggests the hypothesis of a qualia syntax—although not naming it in this way—is Mailman's (2015) article concerning an analytical method he calls Representational Hierarchy Association. With it, the author successfully reduces Schoenberg's Op. 19, No. 3 into a succession of harmonic qualities—diatonicity, hexatonicity, octatonicity, hexatonicity, and diatonicity, respectively—and speculates about the possibility of such qualities play roles in the work's long-range form.

[3] In this regard, it is worth noting how Boss (2019, pp. 224-248) also reduced Schoenberg's Op. 19, Nos. 2 and 3 into progressions of harmonic qualities—in the author's nomenclature, progressions of collections. As such, investigating the hypothesis of a qualia syntax assumes increased importance even in atonal repertoires.

tone quality (Bernardes et al., 2023; Chiu, 2021; Harding, 2021; Quinn, 2007; Yust, 2016, 2019, 2020; Yust et al., 2022).[4] This association emerges from the DFT coefficients' cyclic properties, each capturing unique intervallic structures. Thus, in investigating qualia syntax, this single mathematical framework provides access to various harmonic qualities, enabling comprehension across broad spans of musical history and compositional practices. Furthermore, recent years have witnessed the DFT of pcsets serving as theoretical foundation for geometric interpretations of harmonic-quality relationships (Bernardes et al., 2023; Pereira et al., 2024; Pereira et al., 2025). The Fourier Qualia Space (FQS) (Pereira et al., 2025), a hexagonal space inspired by Quinn's (2006, 2007) quality space, represents one such interpretation, where a pcset's position indicates its qualia characteristics. The concept of harmonic qualia that underpins this space is not merely a relabelling: by unifying Quinn's spatial-geometric treatment of chord quality with Tymoczko's (2011) temporal-accumulative notion of macroharmony, it designates the qualitative properties of harmonic sonorities as measurable positions within mathematically defined spaces, where what a listener experiences as diatonicity or octatonicity corresponds to quantifiable geometric relationships rather than ineffable subjective impressions. The geometric legibility of the FQS stems from each vertex resonating perfectly with one DFT non-trivial coefficient, while central positioning indicates blurred qualia properties—manifesting multiple qualities of equal or approximate intensity. This phenomenon, termed qualia ambiguity, measures the degree to which a pcset or pcset group resonates with DFT qualities (Pereira et al., 2024; Pereira et al., 2025).

Building upon this background, we advance three key contributions. Based on the FQS's partition into six distinct regions, we propose (1) the definition of a new region that establishes qualia ambiguity as a distinct analytical category, effectively treating it as a seventh harmonic quality. To systematically evaluate composers' deployment of harmonic qualia, we introduce (2) a novel computational methodology derived from the FQS's geometric properties for adaptively segmenting works and extracting qualia progressions. Finally, to examine syntax-driven structures in these progressions, we advance (3) a four-tiered nested analysis encompassing a diachronic Zipfian survey spanning nearly four centuries, a synchronic hierarchical perspective on Impressionist music, a detailed transitional-probability study of Debussy's qualia deployment culminating in a three-role model, and work-specific analytical validation.

Regarding the first contribution, the ambiguity region is defined as a circle of radius $\rho = \frac{1}{6}$ centred at the FQS origin; outside this boundary, at least one DFT coefficient necessarily exerts above-average influence on a pcset's position, signifying a predominant quality, whereas points within it reflect balanced magnitude profiles suited to capturing the qualia indeterminacy characteristic of much post-1900 music. The computational pipeline underlying the second contribution operates on the Yale-Classical Archives Corpus (YCAC) (White & Quinn, 2016): frequency-based pitch-class vectors are projected onto the FQS, and segment boundaries are identified through distance-sensitive peak detection on the resulting spatial trajectory; pitch content within

[4] Please note that the terminology varies across the literature since the DFT coefficients measure subtly different qualities according on their phase. For example, the third non-trivial DFT coefficient indicates triadicity in Yust (2019), whereas Chiu (2021) interprets it as a measure of hexatonicity. We opt to adopt the nomenclature reported in Pereira et al. (2025) that reflects the cyclic nature of DFT coefficients.

each segment is then aggregated, logarithmically weighted to attenuate the disproportionate influence of frequently recurring pitch classes, and subjected to a second DFT analysis whose coefficients are classified into the seven qualia regions. As for the four-tiered analysis, log-rank versus log-frequency plots identify periods whose qualia distributions approach a Zipfian slope of −1 (Zipf, 2012, pp. 19–55); hierarchical cluster analysis (Johnson, 1967; Ward, 1963) then reveals how transitional similarities group the seven qualities into dendrograms whose branching patterns disclose functional affinities; first-order Markov transition matrices quantify the directed probabilistic relationships between successive qualia; and two case studies—Debussy's *Mouvement* and *Reflets dans l'eau* (both from *Images*, L. 110; Lesure, 1977)—test the resulting model against the formal architecture of individual works.

The article proceeds as follows: Section 2 draws historical reflections about the concept of musical syntax; Section 3 explicates the DFT of pcsets and related theoretical concepts including the FQS, Zipfian distributions, and Markov chains; Section 4 details our methodological approach for deriving composers' qualia progressions; Section 5 presents statistical and analytical observations within our four-tiered structure; and Section 6 concludes with final remarks and discussion based on our findings.

2. Musical syntax: an overview of its concept and principles

Both musical and linguistic syntax focus on studying the principles that govern the hierarchical organisation of discrete elements into meaningful sequences (Rohrmeier & Pearce, 2018). Despite sharing this definition, musical and linguistic syntaxes fundamentally differ in their building blocks, implying that, unlike language, music lacks explicit semantic content and propositional meaning (Lerdahl, 2013). Given this unique nature of musical discourse, we present in this section a brief literature review of the concept of musical syntax through the centuries, its fundamental principles, and the challenges in post-1900 music syntax.

2.1 Foundations, empirical findings, and the qualia gap

The study of harmonic syntax has a long theoretical lineage—from Rameau's (1971) fundamental bass through Riemann's (1896, pp. 1–9) functional categories (tonic, dominant, subdominant) to Schenker's hierarchical reductions (Schenker & Oster, 2001, pp. 10–117) and Lerdahl and Jackendoff's (1990, pp. 8–11) generative formalism—yet all of these frameworks operate at the level of chord identity and voice-leading function (Lerdahl, 2013; Rohrmeier, 2011; Rohrmeier & Pearce, 2018). Their shared premise is that syntax emerges from the roles individual chords play within key-based hierarchies, a premise that has proven remarkably productive for common-practice tonality but that leaves an entire dimension of harmonic organisation unaddressed: the sequential behaviour of harmonic qualities such as diatonicity or octatonicity.

More recently, computational and statistical approaches have extended the empirical reach of syntactic inquiry. Temperley's (2007a, pp. 23–62) probabilistic models and multiple-viewpoint frameworks (Conklin & Anagnostopoulou, 2001; Conklin & Witten, 1995) demonstrated that harmonic progressions can be modelled as stochastic processes, whilst Rohrmeier and Cross (2008) showed that chord-transition frequencies in Bach's chorales adhere to Zipfian distributions and exhibit marked asymmetry—specific progressions proving far

Table 1. The table presents frequency distributions of the 5 most frequent chord triads in major and minor modes ranked by occurrence in Bach's chorales after transposition for C major and C minor. Data presented by Rohrmeier & Cross (2008).

	Major		Minor	
Rank	chord	count	chord	count
1	{C,E,G}	3217	{C,E♭,G}	2293
2	{D,G,B}	2068	{E♭,G,B♭}	1305
3	{C,F,A}	990	{D,G,B}	1249
4	{C,E,A}	902	{D,F,B♭}	958
5	{D,F,A}	627	{C,F,A♭}	769
6	{D,F,G,B}	383	{C,E♭,A♭}	523
7	{D,F♯,A}	316	{D,G,B♭}	456
8	{E,G,B}	314	{C,E,G}	336
9	{C,D,F♯,A}	210	{C,F,A}	316
10	{E,G♯,B}	201	{D,F,G,B}	222
⋮	⋮	⋮	⋮	⋮

more prevalent in one direction than the reverse, corroborating the hypothesis of goal-directed harmonic motion (see also Huron, 2006, pp. 73–90; Moss et al., 2019; Zanette, 2004). To illustrate this principle, Table 1 and Figure 2 depart from Rohrmeier and Cross's (2008) data, presenting frequency distributions of the 35 most frequent chord triads in major and minor modes ranked by occurrence in Bach's chorales after transposition for C major and C minor, and complementarily depict a similar and strong frequency decay in major and minor chorale chord usage, based on their near-equivalent regression-line slopes ($\alpha = -1.47$ for major and $\alpha = -1.46$ for minor).[5] These findings confirm that musical syntax, like linguistic syntax, organises its elements hierarchically: a small number of structures carry disproportionate structural weight whilst others serve supplementary or connective functions (Lerdahl, 2013; Rohrmeier, 2011).

Crucially, however, all of the above investigations concern chord-level or pitch-level syntax. Whether an analogous organisational logic governs the succession of harmonic *qualities*—the trajectory from, say, diatonic passages through whole-tone regions to octatonic episodes—remains an open question.

2.3 Challenges concerning post-1900 music syntactical frameworks

The analysis and comprehension of musical syntax within post-1900 compositions present substantial challenges in comparison to the conventions of common-practice tonality. The syntactic music frameworks prior to 1900 exhibited a relatively high degree of systematic analysis and prediction. In contrast, the syntactic structures of music created after 1900 display considerable diversity (Forte, 1980, p. ix; Morgan, 1991). Such difference is readily apparent when comparing analytical methodologies applied to pre- and post-1900 musical works (Dunsby, 1993, p. ix). The analysis of common-practice tonality uses established theoretical frameworks, such as Riemannian and Schenkerian analysis, thus facilitating the examination of harmonic

[5] Please note that the slopes mentioned for major and minor modes derive from data in Rohrmeier & Cross (2008) referring only to the 35 most frequent chord structures in Bach's chorales. Performing a similar analysis with the full frequency distribution would probably result in regression slopes that come even closer to $\alpha = -1$, though this remains to be verified.

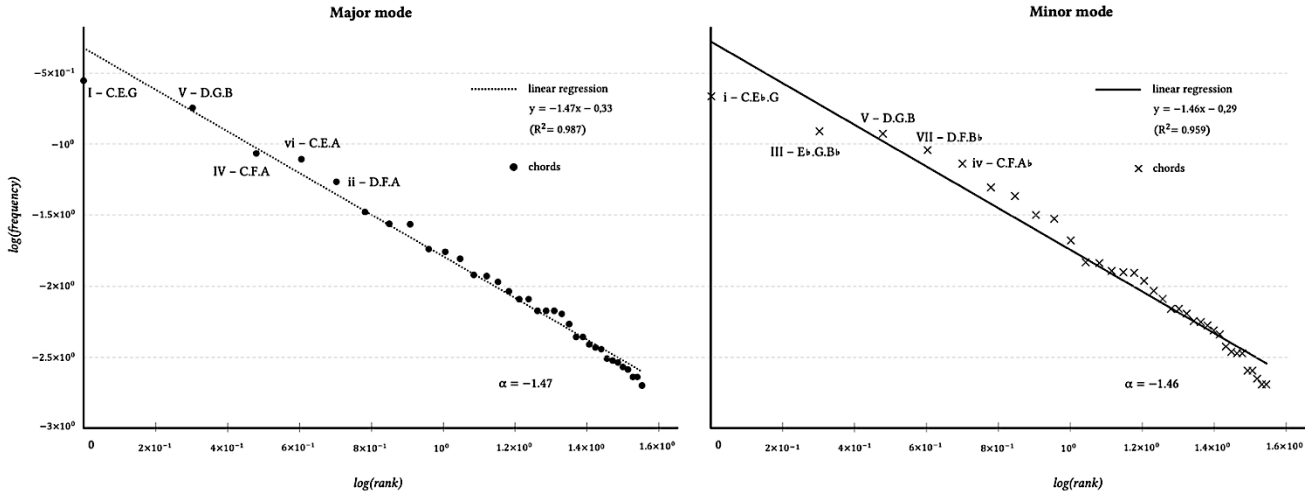


Figure 2. Frequency distribution (log-rank vs. log-frequency) of chord usage in Bach's chorales between major and minor modes. Data extracted from Rohrmeier and Cross (2008) containing the 35 most frequently occurring chords after transposition for C major and C minor. There is a notable similarity in how log-frequencies decrease across both modes, as evidenced by the similar slopes on the regression lines ($R^2 = .987$ for major and $R^2 = .959$ for minor chords).

progression and voice leading, guided by clear principles. In contrast, post-1900 music often resists such systematic analysis, necessitating multiple theoretical approaches to address the disparate compositional styles and techniques that emerge in this musical era (Lerdahl, 2013).

Moreover, one challenging aspect of post-1900 musical analysis is understanding how different harmonic qualities, such as diatonicity and whole-tone quality, interact with and succeed each other (Mailman, 2015). Although Rohrmeier and Moss (2021) have put forth theoretical models for examining these connections through the prism of the *Tonnetz* and its three primary structures—octatonic and hexatonic collections and stacks of fifths—our comprehension of the ways in which these qualities coalesce remains imperfect. In this respect, Somer's (2005) work helps the understanding of this phenomenon by illustrating how composers such as Debussy incorporated these diverse harmonic qualities into coherent syntactic structures, particularly in the composer's late chamber sonatas. His analytical approach demonstrates the feasibility of integrating novel harmonic qualities within traditional phrase structures, thereby suggesting that post-1900 syntax represents a gradual evolution rather than a complete rupture from earlier practices. In this regard, the DFT of pcsets constitutes a significant leap that facilitates the analysis of such evolution, justifying its subsequent detailing.

3. Theoretical and mathematical concepts on Fourier space, Markov, Zipf, and Ward

The analysis of musical harmony has benefited significantly from the application of mathematical and statistical methods. This Section explores four fundamental analytical approaches: Fourier analysis of pcsets (Lewin, 1959, 2001), geometric representation of harmonic qualities through the FQS (Pereira et al., 2025), statistical analysis using Markov chains and Zipf distributions (Temperley, 2007a, pp. 7-22; Zipf, 2012, pp. 19-

Table 2. The established association between the six non-trivial Fourier coefficients ($\hat{f}_1, \ldots, \hat{f}_6$) with their respective harmonic quality, interval class, and specific cyclical features (Amiot, 2016, pp. 91-133; Chiu, 2021; Yust, 2016).

DFT coefficient	Harmonic quality	Interval class	Specific features
$\hat{f}_1$	Chromaticity	1	Examines how the pcset distributes itself across a section of the pitch-class circle.
$\hat{f}_2$	Dyadicity	6	Shows higher magnitude when a pcset contains three or fewer chromatic tritone intervals.
$\hat{f}_3$	Triadicity	4	Measures if a pcset matches or resembles augmented triads or hexatonic collections.
$\hat{f}_4$	Octatonicity	3	Yields increased values for pcsets aligning with diminished seventh chords or octatonic scales.
$\hat{f}_5$	Diatonicity	5	Measures the degree of the pcset's structural organisation within a portion of the circle of fifths.
$\hat{f}_6$	Whole-tone	2	Measures the extent to which the pcset corresponds to or aligns with a whole-tone scale.

55), and Hierarchical Clustering Analysis (HCA; Johnson, 1967; Ward, 1963).[6] These complementary methods provide distinct, yet interconnected perspectives on harmonic structure and progressions, enabling the comprehensive analysis of musical compositions at the heart of our study.

3.1 The Fourier transform of pcsets

Musical set theory provides the mathematical foundation for modern harmonic analysis (Forte, 1980). The theory introduces pcsets and their normal form representations, enabling systematic comparison of harmonic structures. This framework lays the foundation for the application of the DFT of pcsets that reveals geometric properties of harmonic structures through the transformation:

$$\hat{f}_k = \sum_{n=0}^{11} f(n) e^{-2\pi i k n/12} ,$$

where $f(n)$ represents the pitch-class function (with n ranging from 0 to 11 representing the twelve pitch classes) and $\hat{f}_k$ the k-th DFT coefficient (Amiot, 2016, pp. 1-26; Lewin, 1959, 2001). This mathematical approach has proven particularly effective in identifying symmetries and structural relationships in harmonic progressions, offering insights into compositional techniques and stylistic characteristics (Amiot, 2017; Bernardes et al., 2016; Bernardes et al., 2023; Quinn, 2006, 2007; Pereira et al., 2024; Pereira et al., 2025; Yust, 2015a, 2015b, 2016, 2019).

Especially relevant to our work is the intrinsic relationship between the six DFT non-trivial coefficients and six prototypical harmonic qualities, as this connection holds significant music-theoretical relevance (Amiot, 2016, pp. 91-133; Chiu, 2021; Quinn, 2007; Yust, 2016). These coefficients can be mapped to specific structural harmonic qualities in the following manner: DFT coefficient $\hat{f}_1$ quantifies chromaticity, $\hat{f}_2$ measures dyadicity,

[6] Examples of spatial frameworks for musical phenomena include Martins's (2015) dasian space for scale relations in Bartók's music and his expansion of Lutosławski's chordal space (Martins, 2023), the latter proven effective on the harmonic level and in its relationship with form. For comprehensive surveys of geometrical and spatial interpretations of musical structures, see Hook (2023), Krumhansl (2005), and Tymoczko (2011).

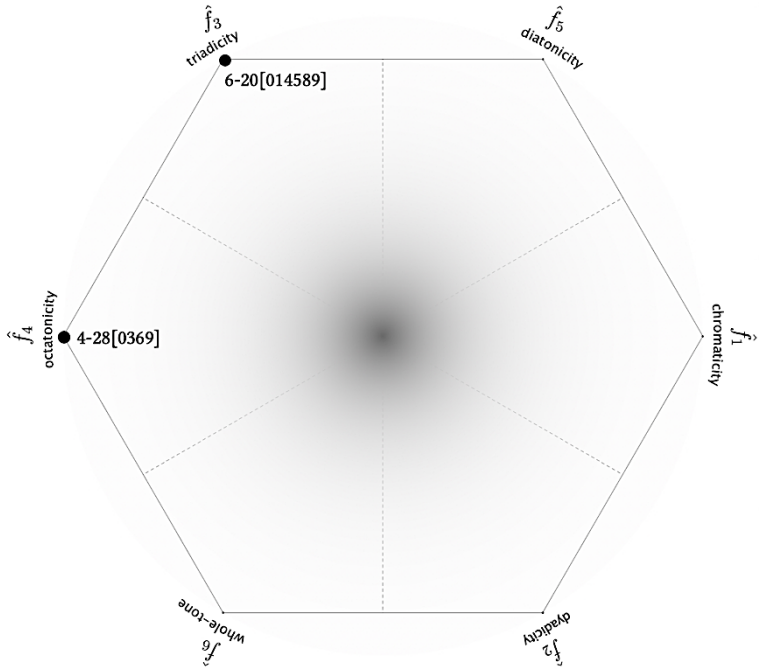


Figure 3. Depiction of the FQS with its six distict regions linked with a harmonic quality (Pereira et al., 2025). The gradient towards the centre illustrates the concept of qualia ambiguity. As contextualising examples, set classes 6-20[014589] and 4-28[0369] appear positioned precisely at the anchors of the $\hat{f}_3$ and $\hat{f}_4$ coefficients as optimal representatives of triadicity and octatonicity.

$\hat{f}_3$ indicates triadicity, DFT coefficient $\hat{f}_4$ represents octatonicity, $\hat{f}_5$ correlates with diatonicity, and $\hat{f}_6$ determines whole-tone quality. Table 2 details the intrinsic characteristics of the relationship between DFT coefficients and harmonic qualities. When analysing a given pitch-class distribution, the magnitude of each DFT coefficient $\left|\hat{f}_k\right|$ serves as a quantitative measure of the distribution's correlation with its corresponding prototypical structure (Amiot, 2016, pp. 91-133; Quinn, 2007). This correlation provides a mathematical framework for assessing the presence and strength of these fundamental musical qualities within the analysed pcset.

3.2 The FQS framework

Built upon the DFT of pcsets framework, the FQS (depicted in Figure 3) is a geometric construct that projects pcsets as points within a hexagon, with each vertex representing one of the six DFT harmonic qualities. In short, it condenses the six-dimensional nature of the DFT's quality space into a comprehensive and musical-meaningful two-dimensional representation. The space utilises the *RadViz* (Inselberg & Dimsdale, 1987) multidimensional data visualisation technique as described in Pereira et al. (2025), and the specific ordering of DFT coefficients ($\hat{f}_1$, $\hat{f}_5$, $\hat{f}_3$, $\hat{f}_4$, $\hat{f}_6$, $\hat{f}_2$) was established through statistical analysis to maximise pcset dispersion while preserving correlations with six-dimensional space. In the FQS, certain pcsets, such as the magic hexachord (014589) and the diminished 7th chord (0369), align perfectly with $\hat{f}_3$ and $\hat{f}_4$ anchors due to

their high saliency in particular DFT coefficients, exactly as Figure 3 shows.[7] Other pcsets demonstrate more complex relationships, often showing affinity to multiple harmonic qualities, as evidenced by their positioning between anchors.

Thus, the FQS allows the assessment of ambiguity, particularly near the centre of the hexagon, since it represents maximum qualia ambiguity, i.e., a positioning in space where multiple harmonic qualities exist with equal intensity. Pereira et al. (2025) also divide the FQS in six regions that represent areas of gradual transition between qualities, with increasing ambiguity towards the centre, leveraging the identification of (1) a bipartite, symmetrical, and consequential structure and (2) ambiguity as an element of Kramer's (1988, 170-83) nondirected linearity and counterpoint in Schoenberg's Op. 19, No.1 and for establishing a hierarchical framework of a work's qualia known as qualia wavescapes (Harasim et al., 2022; Pereira et al., 2024; Viaccoz et al., 2023).

3.3 Statistical approaches to musical analysis: Markov chains, the Zipf distribution, and HCA

We employ first-order Markov chain analysis to model the probabilistic relationships governing successive harmonic events (Pearce & Rohrmeier, 2018; Rohrmeier & Graepel, 2012; Temperley, 2007, pp. 7–22). For an m-th order model, the probability of a harmony h_m depends on the m previous harmonies, denoted $P(h_m \mid h_{m-1}, h_{m-2}, \dots, h_1)$. Two complementary first-order matrices capture different facets of these transitions. A global matrix records the corpus-wide distribution, where each element $G^1_{i,j} = (n_{i\to j}/N) \times 100$ represents the percentage of all observed transitions from state i to state j. A conditional matrix provides a proper Markov transition-probability matrix, where each element $M^1_{i,j} = n_{i\to j} / \sum_k n_{i\to k} \times 100 = P(Q_{t+1} = j \mid Q_t = i) \times 100$ quantifies the probability of reaching state j given the current state i; since each row sums to unity, every state's transitional profile is weighted equally regardless of its overall frequency.

Furthermore, Zipf's law identifies a pattern remarkably robust across different languages and text corpora, reflecting what George Zipf (2012, pp. 19–55) theorised as a least-effort principle—the tension between speakers economising through frequent use of limited vocabulary and listeners benefiting from extensive vocabulary that facilitates precise communication. This relationship finds mathematical expression as:

$$f(z) \approx \frac{\kappa}{z^{\alpha}} \ ,$$

where z represents the rank of a linguistic or musical object, $f(z)$ its occurrence frequency, κ a normalisation constant, and α typically approaches unity. When $\alpha = 1$ precisely, Zipf's law proper emerges; the broader category of Zipf-like distributions encompasses power-law relationships where α may deviate from unity, appearing as straight lines with slope $-\alpha$ on logarithmic plots. Applying Zipfian analysis to musical structures

[7] Set class 6-20[014589], commonly referred to as the magic hexachord, exhibits salience in $\hat{f}_3$ due to its distinctive structure comprising two augmented triads separated by a minor second interval. This configuration perfectly aligns with the position of the $\hat{f}_3$ anchor and serves as a quintessential exemplar of triadic organisation in pitch-class space.

constitutes a logical theoretical development, considering music's nature as an organised communicative system. In addition to Rohrmeier and Cross's (2008) work discussed in Section 2.1, the musical application of this framework emerged decisively through Voss and Clarke's (1975; 1978) pioneering investigations, which demonstrated that pitch and loudness fluctuations across classical, jazz, blues, and rock recordings consistently followed Zipf's distribution—a finding later characterised by Manaris et al. (2005) as evidence for $1/f$ noise patterns pervading diverse musical styles. Their comprehensive analysis of 220 compositions across multiple genres revealed near-Zipfian distributions with an average slope of -1.2023, a value consistent across measurements of pitch, duration, melodic intervals, and harmonic intervals alike (Mandelbrot, 1953).

Finally, we employ HCA to identify relationships between harmonic qualities based on their transitional behaviours (Anderson & Schutz, 2023; Johnson, 1967; Quinn, 1999; Rohrmeier & Cross, 2008; Ward, 1963). The technique follows an agglomerative approach in which individual observations initially form distinct clusters that are iteratively merged based on similarity, using Ward's (1963) minimum-variance criterion to minimise within-cluster variance whilst maximising between-cluster distinctions. The resulting hierarchical structure is visualised through dendrograms, where vertical positioning indicates the dissimilarity at which clusters merge, horizontal branching reveals cluster membership, and the overall structure exposes relationships at multiple scales of resolution (Rohrmeier & Cross, 2008).

4. A computational method to extract and analyse qualia sequences and progressions

In this section, we present a comprehensive methodology for analysing sequences and progressions of harmonic qualities extracted from musical compositions in the YCAC musical corpus.[8] We combine the DFT of pcsets framework, the FQS, and various statistical approaches to examine the progressions and the transitions between harmonic qualia. Our computational pipeline—depicted in detail in Figure 4—processes musical data through eight crucial steps: (1) pcset extraction, (2) first DFT analysis and FQS mapping, (3) segmentation, (4) weighting, (5) second DFT analysis, (6) segments qualia classification, (7) statistical analysis, and (8) output generation. The complete source code implementing this pipeline is available in the supplementary repository.

4.1 Data preparation and pre-processing

We use the YCAC (White & Quinn, 2016) as our main data source for musical works. For each musical piece, we extract all its pcsets and represent them as frequency-based vectors of length 12, where each position corresponds to a pitch class (0-11). Crucially, rather than employing binary representation, our implementation preserves the frequency of each pitch class within a chord, including repeated notes at different octaves (for instance, a chord consisting of notes C4, E4, G4, and C5 would generate the vector [2,0,0,0,1,0,0,1,0,0,0,0]).

[8] We maintain a terminological distinction between *qualia sequence* and *qualia progression*. The former denotes the full ordered series of qualia labels assigned to each segment—including consecutive repetitions of the same quality—whilst the latter denotes the reduced series obtained after eliminating such repetitions (as described above).

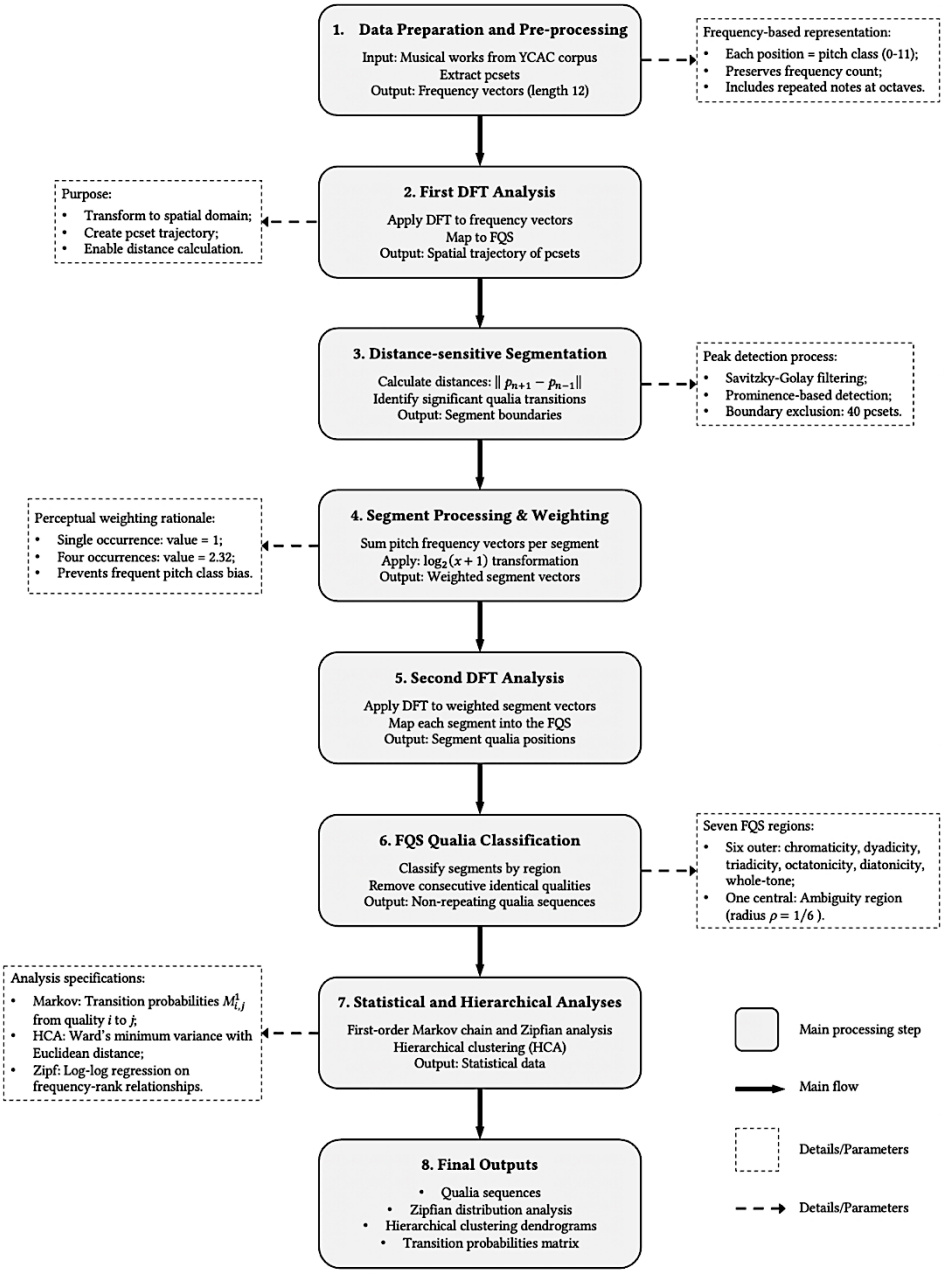


Figure 4. Our computational pipeline for qualia sequence extraction and analysis. Musical data from the YCAC corpus is processed through eight stages: (1) pcset extraction, (2) first DFT analysis and FQS mapping, (3) distance-based segmentation, (4) logarithmic weighting, (5) second DFT analysis for the weighted segments, (6) qualia classification of the segments in the FQS, (7) statistical analyses, and (8) final outputs.

To capture the temporal evolution of harmonic qualities, we implement a distance-sensitive segmentation strategy.[9] This approach identifies significant changes in harmonic qualia by calculating distances between consecutive points in the FQS, thus providing a way to segment works according to their actual qualia

[9] We also performed tests applying a sliding window approach with window size $w = 16$ and interpolation size $\delta = 8$, summing all pitch-classes in a window into a single vector representation. Although this methodology produced results like the ones obtained with the distance-sensitive approach—i.e., identical dendrograms and transition matrices—, we opted for the latter so that we could provide an adaptative segmentation strategy derived from structural changes in a work's harmonic qualia.

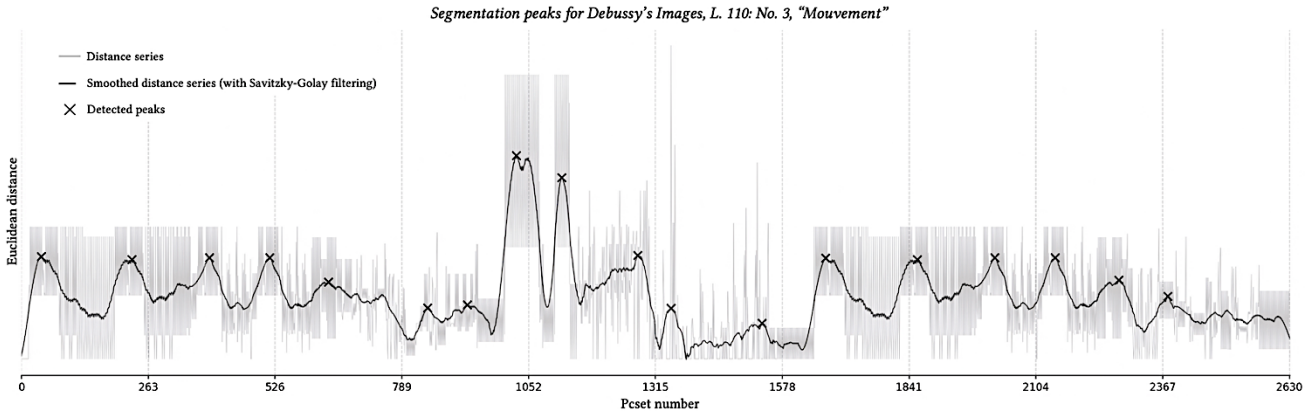


Figure 5. Segmentation procedures for Debussy's *Images*, L. 110: No. 3, *Mouvement*. The grey series represent the Euclidean distance calculated between consecutive pcsets in the FQS, whilst the black curve shows the Savitzky-Golay filtered signal (window length = 65, polynomial order = 3). Crosses (×) mark the detected segmentation boundaries identified through prominence-based peak detection (minimum height = 0.05, minimum distance = 25 pcsets, minimum prominence = 0.05). We also assign boundary regions (40 pcsets from start/end) to prevent edge-effect artefacts.

structure. To this end, we first compute the DFT on the frequency vectors and map them to the FQS, hence obtaining a sequence of pcsets in the space that represent the qualia progression of a work as a spatial trajectory. For each triplet of consecutive points in the FQS (p_{n-1}, p_n, p_{n+1}), we calculate the Euclidean distance between the preceding and following points ($\| p_{n+1} - p_{n-1} \|$), creating a distance series that highlights significant harmonic transitions throughout the piece as graph peaks. As such, we perform peak detection on the resulting distance series using Savitzky-Golay filtering for smoothing, followed by prominence-based peak detection to identify segment boundaries.[10] Additionally, we apply a boundary exclusion threshold of 40 samples (pcsets), whereby peaks detected within this distance from the beginning or end of the sequence are systematically excluded to avoid edge-effect artifacts. This parameterisation mainly serves to identify perceptually significant harmonic qualia transitions as segment boundaries. For a visualisation of the distance-sensitive segmentation strategy main steps, one can inspect in Figure 5 an overlap of the distance series, the resulting Savitzky-Golay smoothed curve, and respective detected peaks for Debussy's *Images, L. 110: No. 3, Mouvement*.

After identifying these segment boundaries, we sum all pitch frequency vectors within each segment to create representative qualia vectors that represent all pitches heard in one segment, also situating our approach within Tymoczko's (2011, pp. 1–19) concept of macroharmony, i.e., the aggregate sonorities emerging from the complete collection of pitches across a moderate to large musical segment. For example, assuming that a given segment contains a C major triad [2, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0], an A minor triad [1, 0, 0, 0, 1, 0, 0, 0, 0, 2, 0, 0], and a G dominant seventh chord [0, 0, 1, 0, 0, 1, 0, 2, 0, 0, 0, 1], the summed qualia vector would be [3, 0, 1, 0, 2, 1, 0, 3, 0, 2, 0, 1]. This aggregate vector reveals that within this segment, the pitches C and G occur

[10] In this regard, we specifically implement the Savitzky-Golay filter with a window length of 65 pcsets (selected to balance local detail preservation with noise reduction), polynomial order of 3, and 'nearest' mode for boundary handling. For the subsequent prominence-based peak detection, we employ the following parameters: minimum peak height of 0.05, minimum distance between peaks of 25 pcsets, minimum peak prominence of 0.05, and minimum peak width of 1.

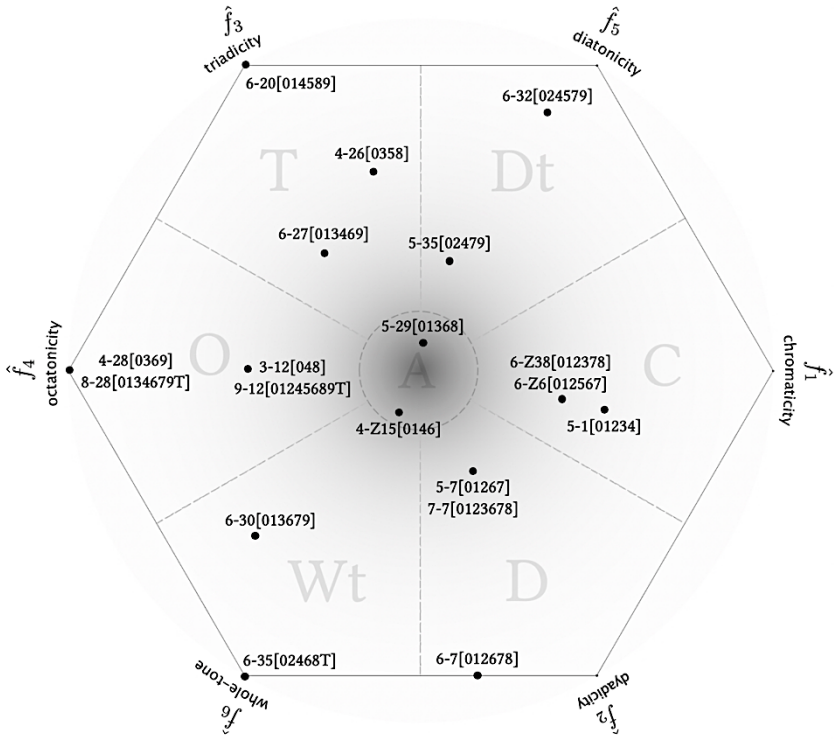


Figure 6. Depiction of the FQS's partition into seven distinct regions thus including the ambiguity zone (A) defined by radius $\rho = 1/6$. In addition, the image shows a set of distinct points representing pcsets and their spatial distribution illustrates how sets are categorised according to their predominant qualia characteristics.

three times, E and A twice, whilst B, D, and F each occur once, effectively capturing the diatonic complexion of the three-chord progression. Moreover, following Chiu (2021) and Harding's (2021) weighting method, we also transform the summed segment vectors with the logarithmic weighting $\log_2(x+1)$. Under this transformation, a pitch class occurring once retains a value of 1, while the transformation modulates a pitch class occurring four times to 2.32, reflecting the perceptual phenomenon wherein the isolated appearance of, for instance, a $\#\hat{4}$ exhibits greater cognitive salience than repeated occurrences of $\hat{1}$ (Chiu, 2021; Harding, 2021). While the general profile of the vector remains intact, the logarithmic function attenuates the impact of higher values, preventing disproportionate influence from frequently recurring pitch classes. Returning to the prior hypothetical summed segment [3, 0, 1, 0, 2, 1, 0, 3, 0, 2, 0, 1], performing the logarithmic weighting to it would result in the following weighted vector [2, 0, 1, 0, 1.585, 1, 0, 2, 0, 1.585, 0, 1]. Subsequently, we employ DFT analysis to these logarithmically weighted segment vectors and map each one onto the FQS.

4.2 Ambiguity region radius definition and qualia classification

Conversely to the FQS's six regions identified in Pereira et al. (2025), we partition it into seven regions, shown in Figure 6. Six outer regions refer to chromaticity, dyadicity, triadicity, octatonicity, diatonicity, and whole-tone quality and one central ambiguity region where pcsets with balanced magnitude saliency lie. Thus, within the qualia sequence, it is possible to integrate segments of the musical structure where there is no predominant harmonic quality.

We adopt the term qualia ambiguity following Temperley's (2007b) definition of tonal ambiguity as "the degree to which a set implies one key as opposed to several"; by extension, qualia ambiguity measures the degree to which a set implies one harmonic quale as opposed to several (Pereira et al., 2025). Since the FQS uses a unit circle in its fundamental structuring, we define the ambiguity region as a circle centred at the origin of the FQS, with its radius defined as $\rho = \frac{1}{6}$. The rationale behind our ambiguity radius threshold choice emerges from the inherent geometrical properties of the FQS, since any point positioned outside the radius ρ, means that at least one DFT coefficient exerts an above-average influence on the point's position, thus indicating a predominant harmonic quality.[11] In contrast, points within this radius necessarily represent pcsets where multiple DFT coefficients contribute in more balanced proportions, as no single coefficient can overcome the combined effect of the others to pull the point beyond the $\rho = \frac{1}{6}$ boundary.[12] As a result, this geometrically driven option partitions the quality space into seven meaningful regions: an inner circle where no single harmonic quality predominates and six outer regions where specific qualities exert a stronger influence. In Figure 6, besides the depiction of the seven distinct regions, it is also possible to inspect the positioning of multiple set classes that lie in each individual region, thus allowing for a comparative view of their intervallic specificities.

After mapping the logarithmically weighted segment vectors into the FQS and classifying them according to the seven-region partitioning schema, we generate a sequence of harmonic qualities for each musical piece—hereafter mathematically noted as C (chromaticity), D (dyadicity), T (triadicity), O (octatonicity), Dt (diatonicity), Wt (whole-tone quality), and A (ambiguity)—such as "$Dt, Dt, O, Wt, \ldots, Dt$". This sequence provides a linear representation of the harmonic qualia unfolding of a work, allowing for a comprehensive study of how such qualia structures behave and interact over time. Moreover, we visualise these sequences of harmonic qualities as simple graphs containing the qualia unfolding and the pcset number for each detected boundary as depicted on Figure 7—the sequence of harmonic qualities obtained when applying our method to Debussy's *Images, L. 110: No. 3, Mouvement*. Finally, after generating the previously mentioned sequence and before conducting transition analysis, we eliminate consecutive identical qualities from it—a methodological step to prevent statistical distortion from sequential redundancies that do not represent qualia changes—to create what we call a qualia progression, also visible in Figure 7. This preprocessing treats a sustained harmonic quality as a single cognitive entity rather than multiple events, thereby producing a more accurate representation of perceptible qualia progressions.

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[11] The ambiguity radius threshold $\rho = \frac{1}{6}$ in the FQS derives from the fundamental geometrical constraints of the *RadViz* technique using six equidistant dimensional anchors on the unit circle. When DFT coefficients contribute uniformly (each with equal magnitude), their vectorial summation situates the point precisely at the origin. Consequently, any point positioned beyond the $\rho = \frac{1}{6}$ threshold necessitates that at least one coefficient exceeds the average contribution, thereby exerting disproportionate influence.

[12] We ran all set classes in the FQS to identify which lie within the $\rho = \frac{1}{6}$ radius ambiguity region due to their balanced DFT coefficient distributions. Among these structures are the monochord 1-1[0], the major third dyad 2-4[04], trichords including 3-2[013], 3-3[014], and 3-6[024], as well as more complex formations such as 4-13[0136] and 5-24[01357], or even highly chromatic sets such as 7-25[0234679], 7-8[0234568], and 12-1[0123456789TE].

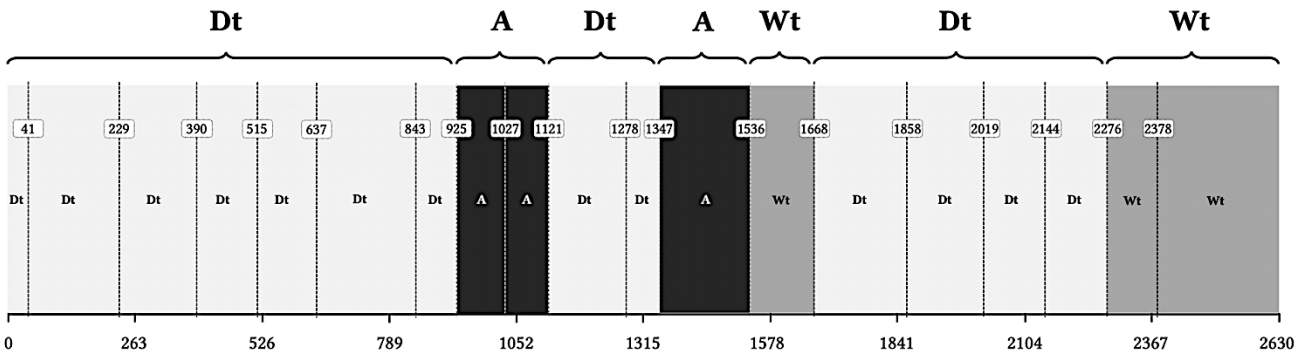


Figure 7. Qualia sequence and progression extracted from Debussy's *Images*, L. 110: No. 3, *Mouvement*. The horizontal axis represents pcset numbers from 0 to 2630, with vertical dashed lines indicating segment boundaries. Each segment's predominant harmonic quality results from FQS mapping and analysis.

4.3 Qualia transition analysis

Subsequently, we perform a first-order Markov chain analysis on the qualia progressions to understand the probabilistic relationships between successive harmonic qualities. Two complementary matrices are computed to capture different aspects of these transitional patterns. The global matrix quantifies the corpus-wide distribution of qualia transitions, where each element $G^1_{i,j}$ represents the percentage of all observed transitions moving from quale i to quale j out of a total transitions, and the conditional matrix provides a proper first-order Markov transition probability matrix, where each element $M^1_{i,j}$ quantifies the probability of transitioning to quale j given the current state is quale i. Since each row of M^1sums to unity, every quale's transitional profile is weighted equally regardless of its overall frequency in the corpus. Together, the two matrices offer complementary perspectives: G^1 captures the absolute prominence of transitional pathways, whilst M^1 isolates each quale's characteristic sequential behaviour independently of its prevalence.

Moreover, we employ HCA to identify relationships between qualia based on their transitional behaviours. Our approach uses Ward's minimum variance method combined with Euclidean distance metrics to create a hierarchical structure that reveals how harmonic qualities relate in terms of their contextual deployment. Finally, we complement the analysis through n-gram identification of recurrent quality patterns and Zipf-like distribution analysis of transition frequencies. The latter examines frequency–rank relationships through log-log regression, enabling quantification of distributional regularities characteristic of complex musical systems.

5. A four-tiered nested analysis of qualia syntax: diachronicity, synchronicity, and Debussy's musical language

Our analytical framework adopts a nested structure, progressively narrowing from broad historical patterns, passing through specific compositional practices, and culminating in specific musical works. This approach encompasses four distinct layers of examination that build upon the results of the previous layer: (1) an investigation into the temporal evolution and emergence of language-like qualia distributions across different musical periods, (2) a comparative analysis of DFT harmonic quality deployment among French school composers, (3) a detailed examination of Debussy's harmonic qualia implementation, culminating in the

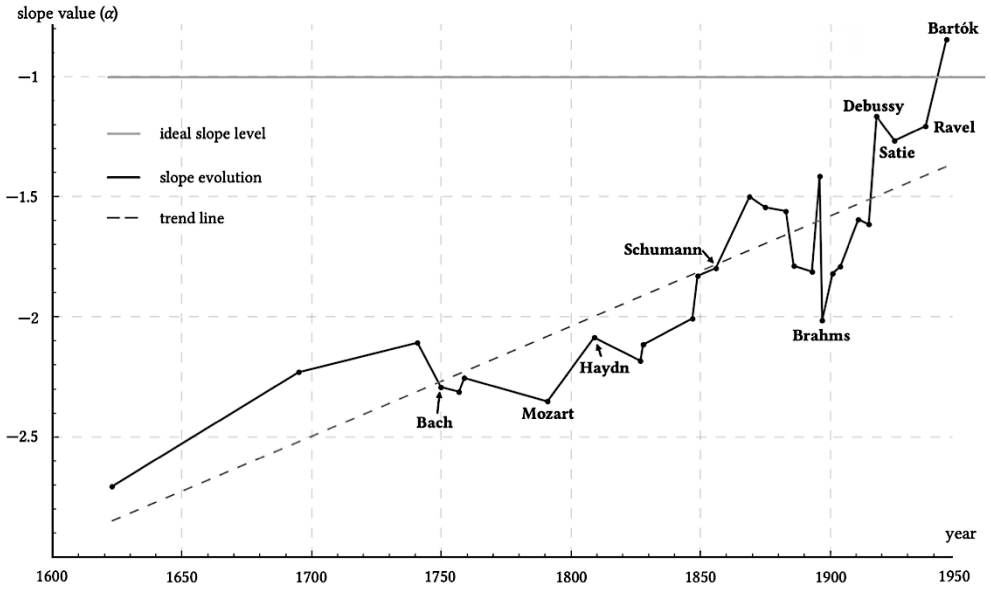


Figure 8. The temporal evolution (1600—1950) of the log-rank/log-frequency regression slopes in qualia progressions exhibits a clear trend (dashed line) towards optimal Zipf's law ($\alpha = -1$, shown as horizontal grey line). The x-axis represents the year of death of the following composers (from the oldest to the most recent year of death): Byrd, Purcell, Vivaldi, Bach, Scarlatti, Handel, Mozart, Haydn, Beethoven, Schubert, Mendelssohn, Chopin, Schumann, Berlioz, Bizet, Wagner, Liszt, Tchaikovsky, Bruckner, Brahms, Verdi, Dvorak, Mahler, Scriabin, Debussy, Satie, Ravel, and Bartók.

systematisation of a three-role model for his compositional practise, and (4) the presentation of two analytical exercises for Debussy's *Mouvement* and *Reflets dans l'eau* (both from *Images*, L. 110) as model's validation.

5.1 First tier: temporal evolution of Zipf-like distributions in qualia progressions

To identify musical periods exhibiting language-like characteristics in qualia and syntactical organisation, we analyse the relationship between log-rank and log-frequency in qualia progressions across multiple composers, spanning approximately 300 years (early-1600s to early-1900s) (Moss et al., 2019; Rohrmeier & Cross, 2008). This analysis serves to identify historical periods where harmonic qualities assume structural significance and potential syntactical structuring in musical composition.

Figure 8 illustrates the temporal evolution of regression line slopes derived from *log*-rank and *log*-frequency plots. The data reveals a progressive trend towards more Zipf-like qualia distributions across centuries, with an approach towards ideal Zipfian slopes emerging at the beginning of the 20th century. Specifically, pre-1800 composers demonstrate steeper regression line slopes between −2.11 and −2.71, whilst 19th-century Romantic composers show moderate slopes between −1.42 and −2.18. The most pronounced Zipfian characteristics appear in post-1900 composers, with regression line slopes ranging from −0.85 to −1.61. Notably, the most Zipf-like qualia progressions appear in works by Bartók (−0.85), Debussy (−1.17), and Ravel (−1.21), whilst the least Zipf-like progressions manifest in Baroque and Classical composers including Byrd (−2.71), Mozart (−2.35), and Bach (−2.31). These findings point to a correlation between Zipf-like slopes and compositional period.

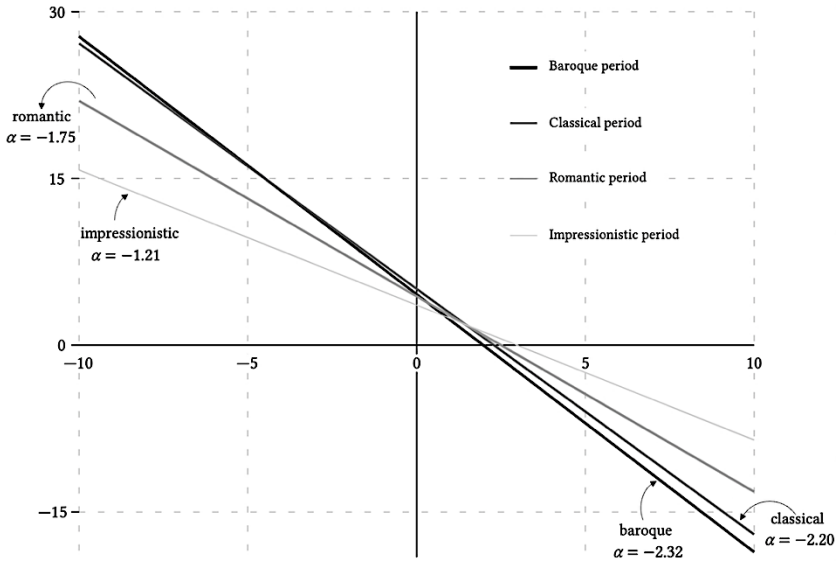


Figure 9. Plot for the mean slope and mean y-intercept values in different musical periods showing slopes approximating -1 through the baroque, classical, romantic, and impressionistic periods. The linear equations result from averages of the folowing composers: (1) baroque period, Byrd, Purcell, Vivaldi, Bach, Scarlatti, and Handel; (2) classical period, Mozart, Haydn, and Beethoven; (3) romantic period, Schubert, Mendelssohn, Chopin, Schumann, Berlioz, Bizet, Wagner, Liszt, Tchaikovsky, Bruckner, Brahms, Verdi, Dvorak, and Mahler; and (4) impressionistic period, Debussy, Satie, and Ravel.

In complement, Figure 9 further supports this correlation through four curves representing mean slope and mean y-intercept values across Baroque, Classical, Romantic, and Impressionist compositional practices. This analysis reveals two significant patterns: firstly, periods associated with tonal functional harmony demonstrate more rigid qualia deployment, particularly through the prevalence of diatonic and triadic transitional structures in both melodic and harmonic contexts, implying a rapid decay in frequency as rank increases. Secondly, modern composers, particularly those of the Impressionist movement, employ qualia progressions that more closely approximate language-like characteristics, i.e., inversely proportional frequencies in relation to its ranks, thus demonstrating a systematic and organised approach to qualia structuring in their compositions.

However, a Zipf-like frequency distribution constitutes a necessary but not sufficient condition for syntactical organisation. Demonstrating syntax requires evidence that elements enter into directed, hierarchical, and functionally differentiated relationships with one another: that certain qualities preferentially precede or follow others, that these preferences cluster into identifiable groups, and that the resulting roles manifest coherently at the level of individual works. The subsequent tiers therefore pursue precisely these dimensions.

5.2 Second tier: A hierarchical perspective of harmonic qualia in Impressionism

To establish a comprehensive hierarchical perspective of qualia treatment in the Impressionist period, we analysed the dendrograms of Debussy and Ravel constructed with the transition probability matrices between the seven qualities and compared them with distinctly different compositional practices through Bach and

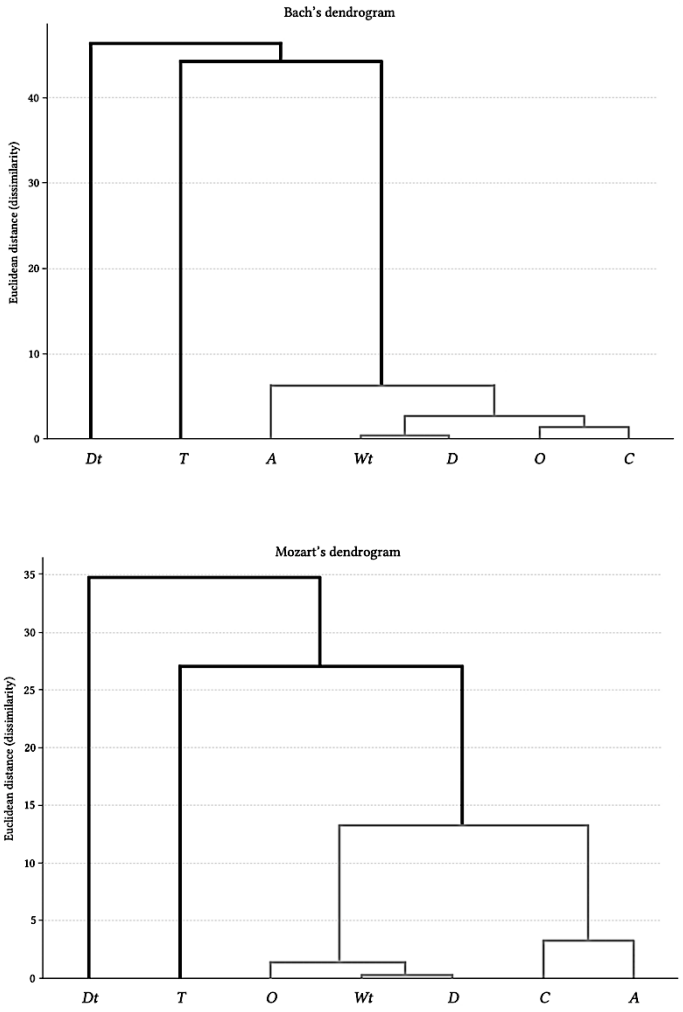


Figure 11. Agglomerative HCA dendrograms for Bach and Mozart's deployment of the DFT harmonic qualities. The y-axis represents relative cluster-centre dissimilarity, whilst the x-axis shows harmonic quality categories. Black and dark grey connections represent the primary and secondary clusters.

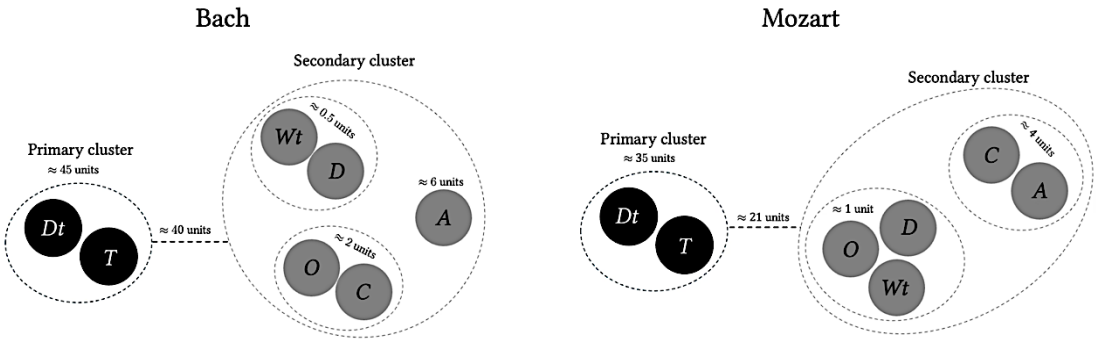


Figure 10. Projected cluster space visualisation of harmonic qualities in Bach and Mozart's compositions. This two-dimensional embedding derives from the respective HCA dendrograms, showing spatial relationships between the DFT harmonic qualities. Primary cluster qualities appear in black whilst secondary cluster qualities appear in dark grey. Distance units indicate relative clustering dissimilarity derived from the original hierarchical analysis.

Mozart's dendrograms.[13] Since these depictions quantify on the y-axis transitional dissimilarity, harmonic qualities separated by greater distances exhibit fundamentally disparate syntactical functions within the composer's compositional framework, while those joined at lower heights demonstrate analogous syntactic

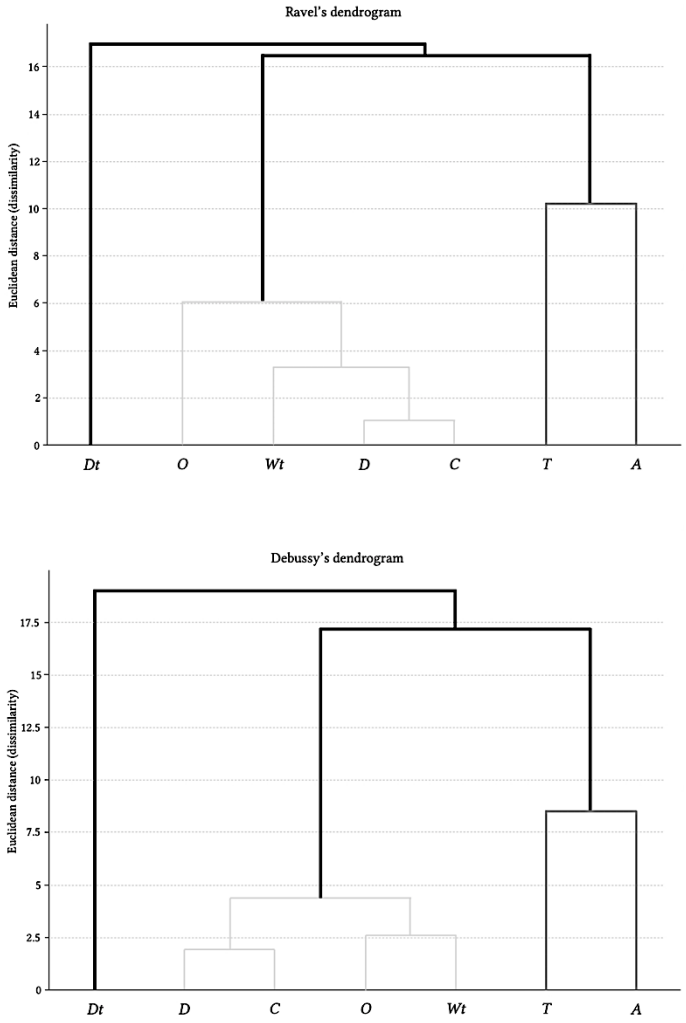


Figure 12. Agglomerative HCA dendrograms for Ravel and Debussy's deployment of the DFT harmonic qualities. Black, dark grey, and light grey connections represent the primary, secondary, and tertiary clusters, respectively.

functions. Complementing the dendrograms, we also present diagrams that briefly describe the clustering space for each composer, offering a more intuitive spatial understanding for transitional pattern dissimilarities as well as enhanced visualisation of cluster and sub-cluster memberships.

The dendrograms of Bach and Mozart in Figure 10 reveal two remarkable consistency points in clustering patterns: (1) at the highest level (in Bach roughly around Euclidean distance 45 and in Mozart at 35) there's a fundamental split between diatonicity and triadicity in the black branch and the remaining qualities in the grey branch that forms a bipartite clustering structure, and (2) the pronounced difference in Euclidean distance between these two main branches reflects how diatonicity and triadicity are profoundly dissimilar in their deployment in Bach and Mozart's repertoires in comparison with the non-diatonic qualities. Figure 11 spatially illustrate these clustering relationships, clearly showing the primary cluster containing *Dt* and *T*

[13] See our supplementary repository to inspect dendrograms referring to the twenty-eight composers in Figure 8 (folder `data/dendrograms/`), their transition count matrices (directory `data/transition_counts/`), and the diachronic Zipfian regression data (`data/diachronic_zipfian_analysis.csv`).

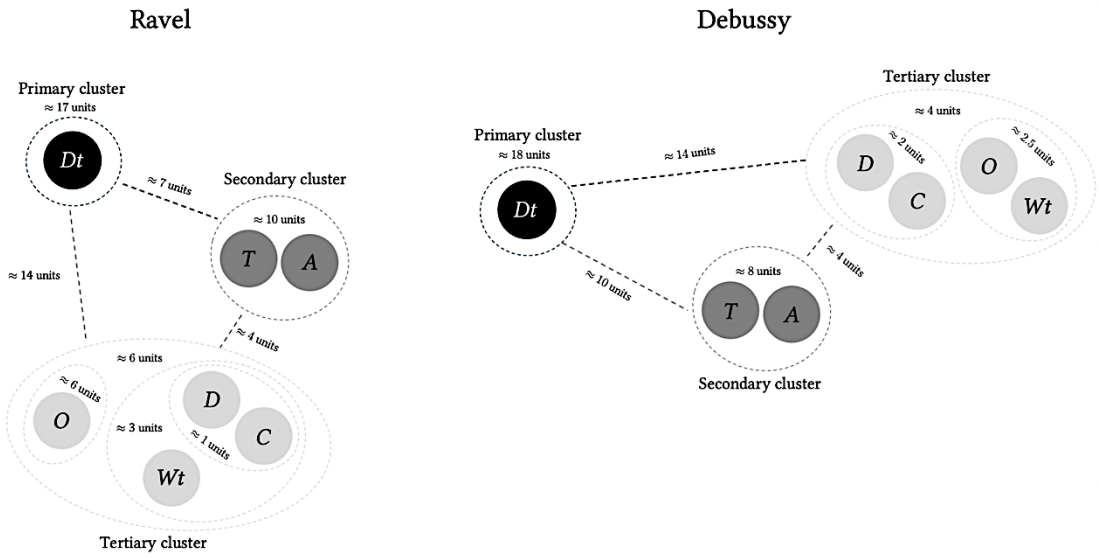


Figure 13. Projected cluster space visualisation of harmonic qualities in Ravel and Debussy compositions. Primary cluster qualities appear in black, secondary cluster qualities in dark grey, whilst tertiary cluster qualities appear in light grey.

as a cohesive unit—functioning almost as a qualia nucleus—, with secondary and tertiary clusters distributed at considerably greater distances and playing peripheral roles.

In noticeable contrast, both Ravel's and Debussy's dendrograms depicted in Figure 12 reveal markedly distinctive characteristics. Their more diverse and unpredictable grouping patterns suggest composer-specific treatments of harmonic qualia, indicating individual harmonic languages. These dendrograms further illustrate two significant departures from functional tonal thinking: firstly, diatonicity appears notably isolated in all composers, though with comparatively reduced Euclidean distances in the Impressionist composers (~17.5 units in Debussy, ~16.5 units in Ravel) relative to their Classical and Baroque counterparts (approximately 45 and 35 units in Bach and Mozart, respectively). This indicates a modified but still distinct functional role for diatonicity within the Impressionist harmonic framework. Secondly, the four modern harmonic qualities—chromaticity, dyadicity, octatonicity, and whole-tone quality—exhibit larger Euclidean distances, implying their role as fundamental building blocks rather than mere decorative elements.

Moreover, unlike Bach and Mozart's bipartite harmonic division, Debussy and Ravel exhibit tripartite hierarchical structures comprising: (1) isolated diatonicity, (2) paired triadic-ambiguous elements—a significant departure from earlier composers where triadicity aligned with diatonicity—and (3) non-diatonic clusters. The spatial representations in Figure 13 demonstrate these tripartite structures, with Ravel showing a more compact organisation and Debussy displaying a more dispersed arrangement with a distinctive tertiary cluster. The internal organisation of these clusters reveals composer-specific signatures: Debussy demonstrates proximal $D - C$ (~2 units) and $O - Wt$ (~2.5 units) pairings, whereas Ravel exhibits integrated $Wt - D - C$ elements (~1-3 units) with O maintaining relative independence (~6 units). These distinctive sub clustering patterns reflect individuated compositional methodologies within a shared revolutionary harmonic framework. Nevertheless, the increased flexibility in incorporating diverse qualities does not indicate arbitrary deployment. Instead, the tripartite hierarchy and the distinct degrees of dissimilarity between each

Table 3. Complete set of the Debussy's analysed works showing compositional chronology, Lesure's (1977) catalogue numbers (L.), slice counts, and extracted qualia progressions. We obtained for each quality the following total occurrence counts: ***Dt*** – 89, ***A*** – 82, ***T*** – 62, ***O*** – 30, ***Wt*** – 28, ***C*** – 13, and ***D*** – 9.

#	Year	L.	Title	Slices	Qualia Progression
1	1880	L. 3	Piano Trio in G major: II. *Scherzo*	1018	*T – Dt – T – Dt – T – Dt*
2	1880	L. 3	Piano Trio in G major: III. *Intermezzo*	1149	*Dt – T*
3	1882	L. 29	*Mandoline* (text by Paul Verlaine)	406	*Dt*
4	1885	L. 79	*Deux romances*: I. *L'âme évaporée*	206	*Dt*
5	1886—1889	L. 65	*Petite suite*: II. *Cortège*	1137	*Dt – A – Dt*
6	1888—1891	L. 66	*Deux Arabesques*: No. 1, *Première Arabesque*	1271	*T – Dt – T – Dt*
7	1888—1891	L. 66	*Deux Arabesques*: No. 2, *Deuxième Arabesque*	1368	*Dt – T – Dt – A – Dt – T*
8	1890	L. 70	*Ballade* (originally *Ballade slave*)	2643	*A – Dt – O – Dt – T – Dt – A – T – Dt – T – Dt*
9	1890	L. 68	*Rêverie*	826	*A – Dt – T – Dt – A – Dt – T*
10	1890—1905	L. 75	*Suite bergamasque*: II. *Menuet*	1148	*A – T – Dt – C – A – T – Dt*
11	1890—1905	L. 75	*Suite bergamasque*: III. *Clair de lune*	809	*Dt – T – Dt*
12	1890—1905	L. 75	*Suite bergamasque*: IV. *Passepied*	1738	*Dt – A – Dt – T – Dt – A – O – T – A – T – A – Dt – A*
13	1892—1894	L. 86	*Prélude à l'après-midi d'un faune*	2227	*O – T – A – Dt – O – D – O – D – O – A – C – T – A – Dt – T – Dt – T – O – Dt*
14	1893	L. 85	String Quartet in G minor, Op. 10	3021	*T – Dt – A – T – D – A – T – Dt – A – Dt – A – Wt – Dt – O – T – Wt – T – A – Dt – A – Dt –*
15	1897—1899	L. 91	*Nocturnes*: I. *Nuages*	1564	*A – Dt – O – A – Dt – Wt – T – Dt – O – Wt*
16	1897—1899	L. 91	*Nocturnes*: III. *Sirènes*	3352	*Dt – T – O – Wt – T – Dt – O – A – Wt – A – O – A – Wt – O – Wt*
17	1898	L. 92	*Trois chansons de Charles d'Orléans*, I. *Di...*	286	*Dt*
18	1901—1905	L. 110	*Images, Livre* 1: I. *Reflets dans l'eau*	1482	*Dt – T – A – O – Wt – A – T – Dt – A – Wt – T – A*
19	1901—1905	L. 110	*Images, Livre* 1: II. *Hommage à Rameau*	642	*Dt – A – Dt*
20	1901—1905	L. 110	*Images, Livre* 1: III. *Mouvement*	2632	*Dt – A – Dt – A – Wt – Dt – Wt*
21	1903	L. 100	*Estampes*: I. *Pagodes*	1730	*Dt*
22	1903	L. 100	*Estampes*: II. *La soirée dans Grenade*	943	*A – Dt – A – C – Dt – T*
23	1903	L. 100	*Estampes*: III. *Jardins sous la pluie*	2557	*T – Dt – A – C – Dt – O – T – A – Wt – T – A – O – A – Wt – A – Wt – Dt – C – Dt*
24	1903—1904	L. 106	*L'isle joyeuse*	3124	*Wt – A – Wt – Dt – T – Wt – Dt – A – Wt – O – Wt – A – Wt – A – Dt – A – Wt*
25	1903—1905	L. 109	*La mer, trois esquisses symphoniques*: I. *De...*	3622	*Dt – T – A – T – Dt – A – T – C – A – O – T – Dt – A – Dt – O – Dt – A – Dt – A – Wt – T – A*
26	1906—1908	L. 113	Children's Corner: II. Jimbo's Lullaby	566	*Dt – C – Wt – O – Dt*
27	1906—1908	L. 113	Children's Corner: V. The Little Shepherd	245	*A – Dt – A*
28	1906—1908	L. 113	Children's Corner: VI. Golliwogg's…	738	*A – Dt – A – Dt – T*
29	1907	L. 111	*Images, Livre* 2: II. *Et la lune descend sur...*	503	*Dt – T – Dt – O*
30	1907	L. 111	*Images, Livre* 2: III. *Poissons d'or*	1817	*A – Dt – T – A – T – Dt – A – Dt – T – Wt – A – Wt – A – T – Dt – O – T – Dt*
31	1909—1910	L. 116	*Première rhapsodie pour clarinette et piano*	3111	*Dt – A – Wt – A – T – C – T – Dt – A – T – A – T – A – Wt – A – Dt – O – C – Dt*
32	1915	L. 136	*Douze Études*: No. 2, *Pour les tierces*	1211	*Dt – T – Dt – T – C – Dt – A*
33	1915	L. 136	*Douze Études*: No. 3, *Pour les quartes*	890	*A – Dt – A – Dt*
34	1915	L. 136	*Douze Études*: No. 5, *Pour les octaves*	968	*Dt – T – Dt – T – Dt – A – O*
35	1915	L. 136	*Douze Études*: No. 7, *Pour les degrés...*	1299	*D – Dt – O – A – D – C – A – D – A*
36	1915	L. 136	*Douze Études*: No. 9, *Pour les notes...*	1532	*O – A – D – A – C – D – O – A – O – D – C – A*
37	1915	L. 136	*Douze Études*: No. 10, *Pour les sonorités...*	590	*O – A – Dt – T – Dt*
38	1915	L. 136	*Douze Études*: No. 11, *Pour les arpèges...*	1136	*A – Dt – T – Dt – A*
39	1915	L. 136	*Douze Études*: No. 12, *Pour les accords*	686	*T – Dt – A – Dt – A – Dt*

harmonic quality within the dendrogram structures support the hypothesis of an underlying syntactical system, albeit one fundamentally different from that of Bach and Mozart.

5.3 Third tier: towards a role-model of qualia relationships in Debussy's music

Our following examination of Debussy's qualia progressions further investigates their syntactical structure. Such progressions—catalogued in Table 3 with the correspondence between piece name, number of slices, and qualia progression—were obtained by applying our segmentation and quality assignment strategy detailed in Section 4. Moreover, to assess the sensitivity of the transition matrices to segmentation parameters, we computed M^1 for all 27 combinations of window length $\in \{45, 65, 85\}$, prominence $\in \{0.03, 0.05, 0.07\}$, and

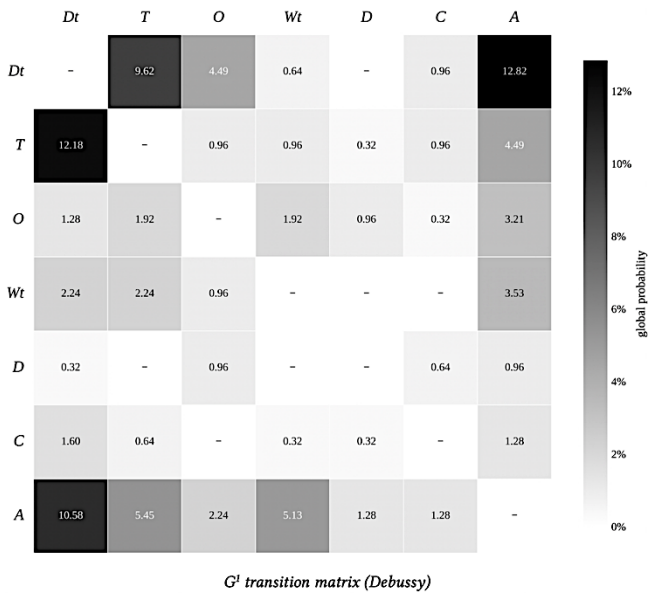


Figure 14. Global transition probability matrix G^{1} for Debussy's qualia sequences. Each cell $G^{1}_{i,j}$ represents the percentage of all observed transitions that move from quale i (row) to quale j (column). Darker shading indicates higher percentages.

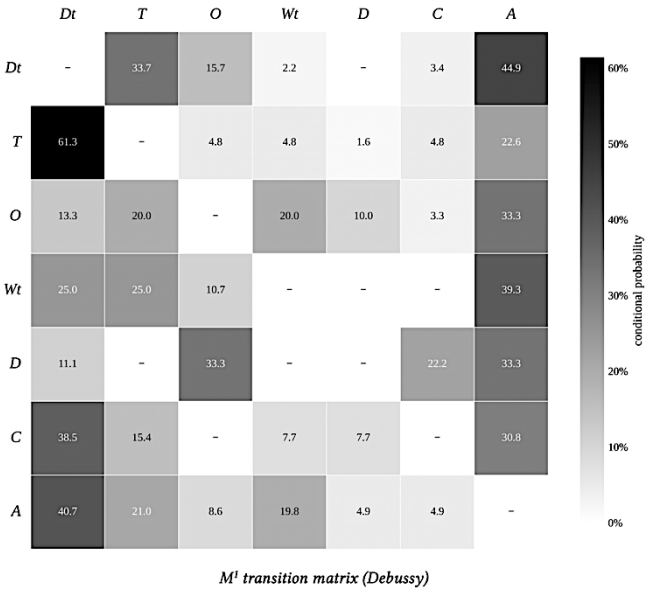


Figure 15. Conditional probability matrix M^{1} for Debussy's qualia unfolding. Each cell $M^{1}_{i,j}$ represents the conditional probability $P(Q_{t+1} = j \mid Q_t = i)$ of transitioning to quale j given the current state is quale i.

minimum peak distance $\in \{15, 25, 35\}$ across Debussy's catalogue. Single-parameter perturbations from the baseline configuration $\{65, 0.05, 25\}$ yielded Pearson correlations of $r > 0.91$ with the baseline matrix, indicating that the following transition structure is robust to local parameter variation. Simultaneous perturbation of all three parameters reduced correlations to $0.8 \lesssim r \gtrsim 0.85$, with window length emerging as the dominant source of variation. In no configuration did the qualitative structure of the matrix—the rank ordering of transition probabilities—collapse. Full results are reported in supplementary repository (see `tests/parameter_sensitivity.py`).

Table 4. Condensed classification of transitions between harmonic qualities in Debussy's works, derived from conditional transition probability data (M^1). Transitions are organised into three frequency bands: *Often followed by* ($\geq$ **30**), *Sometimes followed by* ($\geq$ **10** and $<$ **30**), and *Seldom followed by* ($<$ **10**%).

	Often	Sometimes	Seldom
C	Dt, A	T	Wt, D
D	O, A	C, Dt	–
T	Dt	A	O, Wt, C, D
O	A	T, Wt, Dt, D	C
Dt	A, T	O	C, Wt
Wt	A	Dt, T, O	–
A	Dt	T, Wt	O, D, C

Table 5. A three-layered framework for organising harmonic qualia within Debussy's compositional practice. Drawing on HCA and transition pattern analysis (global and conditional), the model distributes the seven DFT-based harmonic qualities across functional strata.

Layer	Qualities	Role	Characteristics	Transitions
Qualia anchor	Dt	- Stability source - Reference point	- Structural isolation - Initiates progressions	- High outgoing transitions - Strong links to A and T
Mediating transitional	T, A	- Pivot - Flexibility	- Transitional centrality - Layers connection	- Bidirectional with Dt - Bridges to O, Wt, C, D
Colouristic contrast	O, Wt C, D	- Tension, contrast - Colour	- Brief interventions - Qualia contrast	- Primarily connects through mediating layer

The global and conditional transition probability matrices, visualised in Figures 14 and 15 respectively, reveal a structured network of qualia transitions that can be parsed into three functionally distinct groups.[14] Table 4 summarises these intrinsic transitions by identifying qualities that frequently, occasionally, or rarely follow each DFT harmonic quality. Based on these findings and the HCA analysis depicted in Figure 12, we propose a three-layer model describing specific roles for each layer in Debussy's qualia progressions, as summarised in Table 5 and visualised in Figure 16.[15]

The first layer plays a *qualia anchor role*, comprising diatonicity. The global matrix reveals that the bidirectional relationship between diatonicity and triadicity ($Dt \leftrightarrow T$) accounts for 21.79% of total transitions, whilst interactions between diatonicity and ambiguity ($Dt \leftrightarrow A$) account for a further 23.4%. Diatonicity's predominant function as a structural initiator emerges clearly: the transition $Dt \rightarrow A$ constitutes the single most frequent pathway (12.82%), with significantly higher probabilities of departure than arrival. The conditional matrix amplifies this observation: triadicity's 61.3% conditional probability of returning to diatonicity ($M^1_{T,Dt}$) represents the highest value in the entire M^1 matrix, confirming diatonicity's gravitational pull. The pronounced structural isolation in the dendrogram—positioned at approximately 17.5 units

[14] Debussy's global and conditional transition matrices, as well as the complete processed dataframe, are provided in the supplementary repository (`data/debussy/`).

[15] The three role names reflect the transitional behaviour each layer exhibits. *Anchor* captures diatonicity's dual function as both departure point and gravitational return, as the transition probabilities confirm; *mediating transitional* describes how triadicity and ambiguity facilitate movement between harmonic regions through their balanced incoming and outgoing transitions; and *colouristic contrast* reflects the brief appearances and distinctive sonic character that the remaining qualities contribute, providing variety and expressive contrast within the larger harmonic framework.

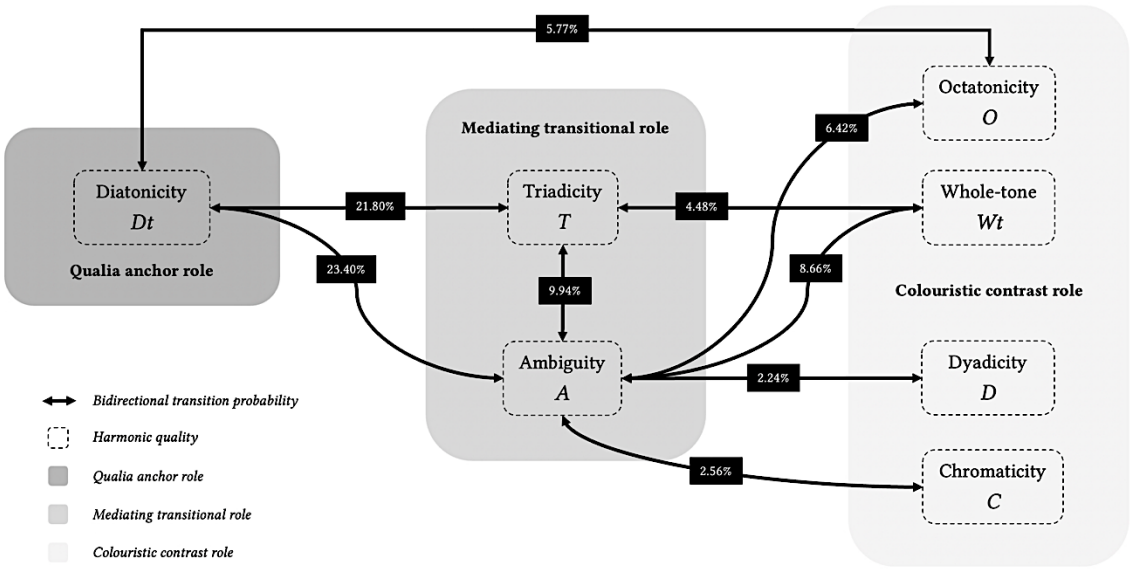


Figure 16. Schematic representation of the three-layer model showing bidirectional transition probabilities between harmonic qualities. Arrow percentages indicate the summed bidirectional transition probabilities, revealing the central mediating function of triadicity and ambiguity in connecting all three layers.

distance—further substantiates this designation. This quality forms the core of Debussy's qualia progressions and manifests itself mainly through modal and pentatonic passages.

The second layer plays a *mediating transitional role*, containing ambiguity and triadicity, which function as pivots facilitating smooth qualia movement. Their clustering together in the dendrogram at approximately 8.5 units—positioned hierarchically between diatonicity and the colouristic qualities—suggests shared functional characteristics despite their theoretical differences. The conditional matrix substantiates this classification: ambiguity's row in M^1 shows the most evenly distributed profile among all qualia, with departures spread across diatonicity (40.7%), triadicity (21.0%), and whole-tone quality (19.8%), whilst triadicity—despite its strong gravitational return to diatonicity—still directs 22.6% of its transitions towards ambiguity. These qualities exhibit demonstrable network centrality and maintain balanced incoming/outgoing transitions, functioning as connective hubs linking all three hierarchical clusters, as outlined in Table 5.

The third layer plays a *colouristic contrast role*, comprising octatonicity, whole-tone quality, chromaticity, and dyadicity. Their remarkably tight clustering at low hierarchical distances (approximately 2–2.5 units) indicates strong functional similarity despite their theoretical divergence. Transition patterns reveal limited direct connections to diatonicity, with these qualities instead routing preferentially through the mediating layer: the conditional matrix confirms that ambiguity receives 33.3% of octatonicity's departures, 39.3% of whole-tone's, 33.3% of dyadicity's, and 30.8% of chromaticity's—consistently the highest or joint-highest conditional value in each row. One notable exception is the direct connection between diatonicity and octatonicity, summing to 5.77% in both directions. Their transitional behaviour demonstrates patterns of brief utilisation rather than extended stretches, functioning as momentary colouristic interventions within larger progressions.

Before proceeding to work-level validation, we dedicate a few lines on assessing the robustness of the three-role classification with respect to the ambiguity radius. To do so, we repeated the qualia assignment for $\rho \in \{\frac{1}{8}, \frac{1}{7}, \frac{1}{5}\}$ alongside the baseline $\rho = \frac{1}{6}$, regenerating occurrence counts and both transition matrices for Debussy's catalogue at each threshold. Diatonicity's occurrence count is virtually invariant across thresholds (86–91), and the conditional probability $M^1_{T,Dt}$ (the chief quantitative signature of its anchor role) never falls below 58.9%. The most visible effect of varying ρ is a redistribution within the mediating layer: a smaller radius reclassifies borderline ambiguous segments as triadic, whilst a larger one absorbs borderline triadic segments into ambiguity. Crucially, however, the combined share of diatonicity's conditional departures directed towards mediating qualities ($M^1_{Dt,T} + M^1_{Dt,A}$) remains consistently high across all thresholds (67.4%–86.8%), and colouristic qualities continue to route their departures preferentially through ambiguity rather than directly to the anchor. We note that $\rho = \frac{1}{5}$ arguably represents the model's upper boundary, as ambiguity at this threshold surpasses diatonicity in count and concentrates departures onto a single pathway. Taken together, these results confirm that the three-role model is a structural property of Debussy's harmonic language, not an artefact of the $\rho = \frac{1}{6}$ mathematical choice. The script `tests/ambiguity_radius.py` provided in supplemental material summarises these test results for $\rho \in \{\frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}\}$.

5.4 Fourth tier: work-specific validation of the model

This fourth and innermost tier tests the three-role model against individual compositions, asking whether the functional roles identified through aggregate transition probabilities manifest coherently across the formal architecture of particular works. We address this question through two case studies from Debussy's *Images*, L. 110: *Mouvement* illustrates the model's standard operation across an entire formal design, whereas *Reflets dans l'eau* discloses finer-grained complexities, including context-dependent qualia classification and the mediating layer's role in enabling transitions between anchor and colouristic passages.

The segmentation and qualia progression of *Mouvement*, shown in Figure 7 and revisited here as Figure 17, broadly traces the harmonic trajectory of the piece from diatonicity through increasing ambiguity towards whole-tone quality, yielding the progression $Dt \to A \to Dt \to A \to Wt \to Dt \to Wt$ that encapsulates the listening experience of the musical discourse.

A defining characteristic of the work is a triplet perpetual motion, shown in Figure 18, that inexorably shapes the underlying texture as an unstoppable *mouvement*. The opening four measures, despite consisting entirely of simple perfect fifths (C and G), deploy exquisite articulation to establish the rhythmic pattern that propels the music forward: a group of four quavers articulated with three staccatos and a final emphasised tenuto (see Figure 18's left depiction). Once the triplet motion commences in m. 5, the pattern expands into a group of seven staccatos and one tenuto, and is further reinforced by brief departures from the diatonic set comprising {C,D,E,F,G} to the octatonic subset {C,E♭,F♯,G,A}, as Figure 18's right panel confirms. These fleeting octatonic allusions are insufficient to produce a clear octatonic flavour for the listener, yet they plainly underscore the

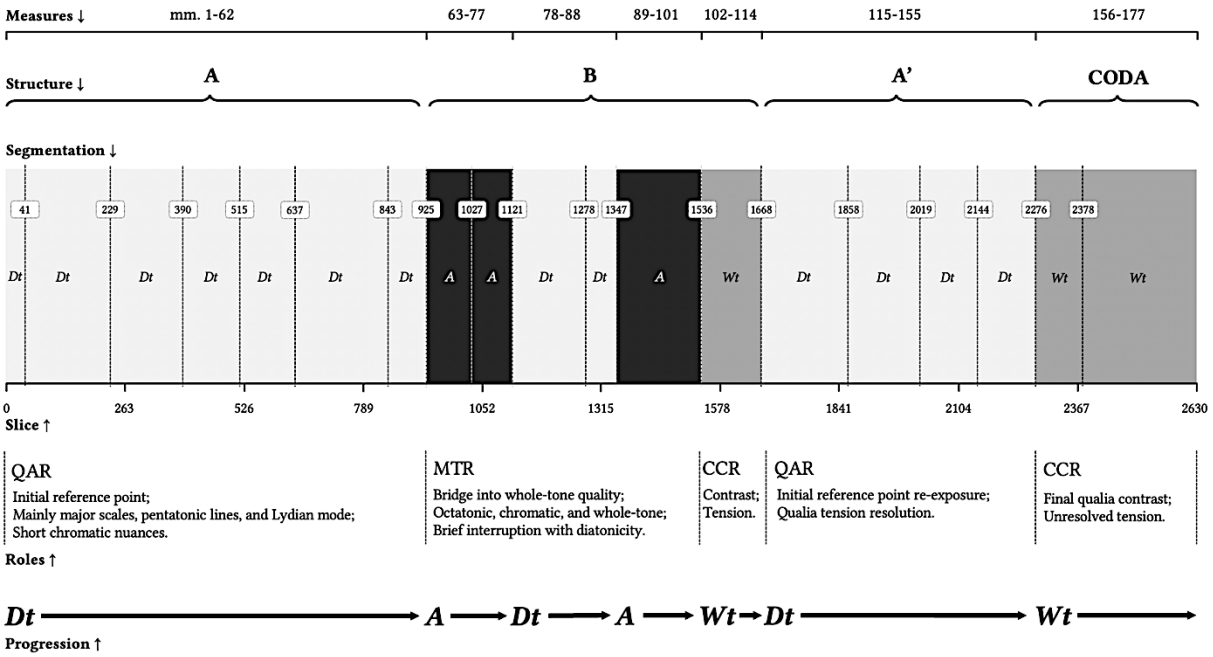


Figure 17. An integrated analysis and segmentation of Debussy's *Images, L. 110: No. 3, Mouvement*, presenting our algorithm's qualia-based segmentation and progression alongside formal design (ABA') and interpretation through the three-role framework (QAR = qualia anchor role, MTR = mediating transitional role, CCR = colouristic contrast role).

articulation pattern at the tenuto. This elegant device—a moving pattern whose final element receives emphasis—assumes still greater preponderance, since the qualia progression itself appears to reinforce it.

The opening diatonic passages (mm. 1–29 and mm. 42–62), together with pentatonic material defining a melody played in both hands (mm. 30–41), establish a firm anchor. Even the octatonic suggestions illustrated in Figure 18 and subsequently extended in mm. 12–15 for the subset {C,E♭,F♯,G,A,B♭} fail to dilute the diatonic character, since five of its six notes form the C Dorian subset {C,E♭,G,A,B♭}. Towards the close, structural diatonicity reappears (mm. 115–155) with the re-emergence of familiar melodic themes after interruptions of ambiguity and brief whole-tone referencing, as Figure 17 depicts.

The ambiguous passages (mm. 63–77 and mm. 89–101) feature heightened chromatic voice-leading alongside multiple concurrent pitch collections, functioning as transitional zones where overlapping diatonic and whole-tone components prepare imminent qualia transformations. The separation of these instances by an

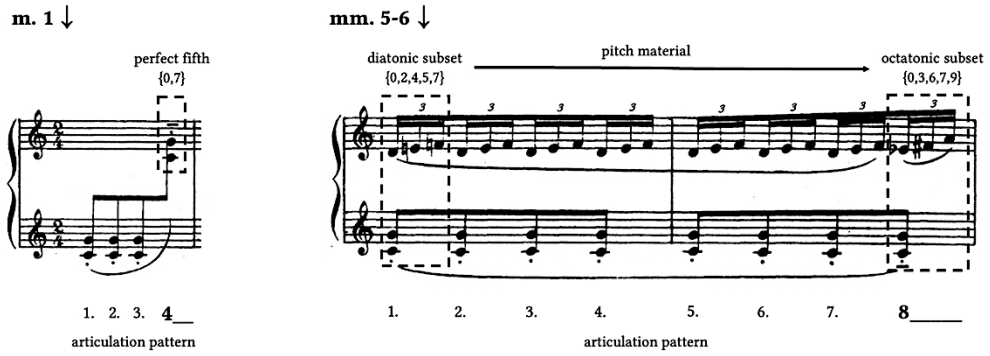


Figure 18. The rhythmic (triplet perpetual motion), articulative (groups of 3+1 and 7+1 eight note), and pitch organisation patterns (diatonic briefly interrupted by octatonic subsets) creating the sense of *mouvement* in Debussy's *Images, L. 110: No. 3.*
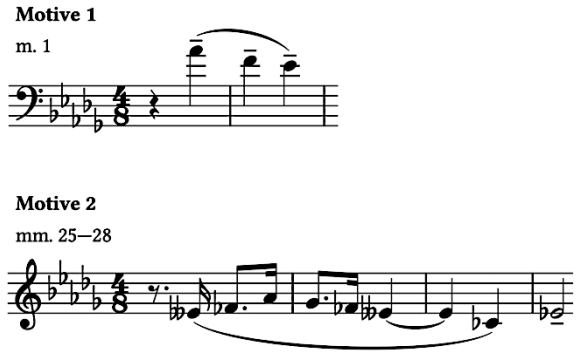


Figure 19. The two melodic motives driving *Reflets dans l'eau*. Both motives articulate critical formal boundaries: Motive 1 marks qualia anchor returns (mm. 1—24, 36—50, and 82—95), while Motive 2 appears at colouristic peaks (mm. 25—35 and 51—71).

intervening diatonic stretch (mm. 78–88) proves especially revealing: ten measures during which the volatile qualia terrain momentarily stabilises.

Figure 17's segmentation additionally discloses two distinct whole-tone emergences. The first (mm. 102–114) concludes section B of the A–B–A′–CODA form, whilst the second (mm. 156–177), constituting the CODA, closes the piece in what might be termed a qualia imperfect cadence—an unresolved colouristic departure that

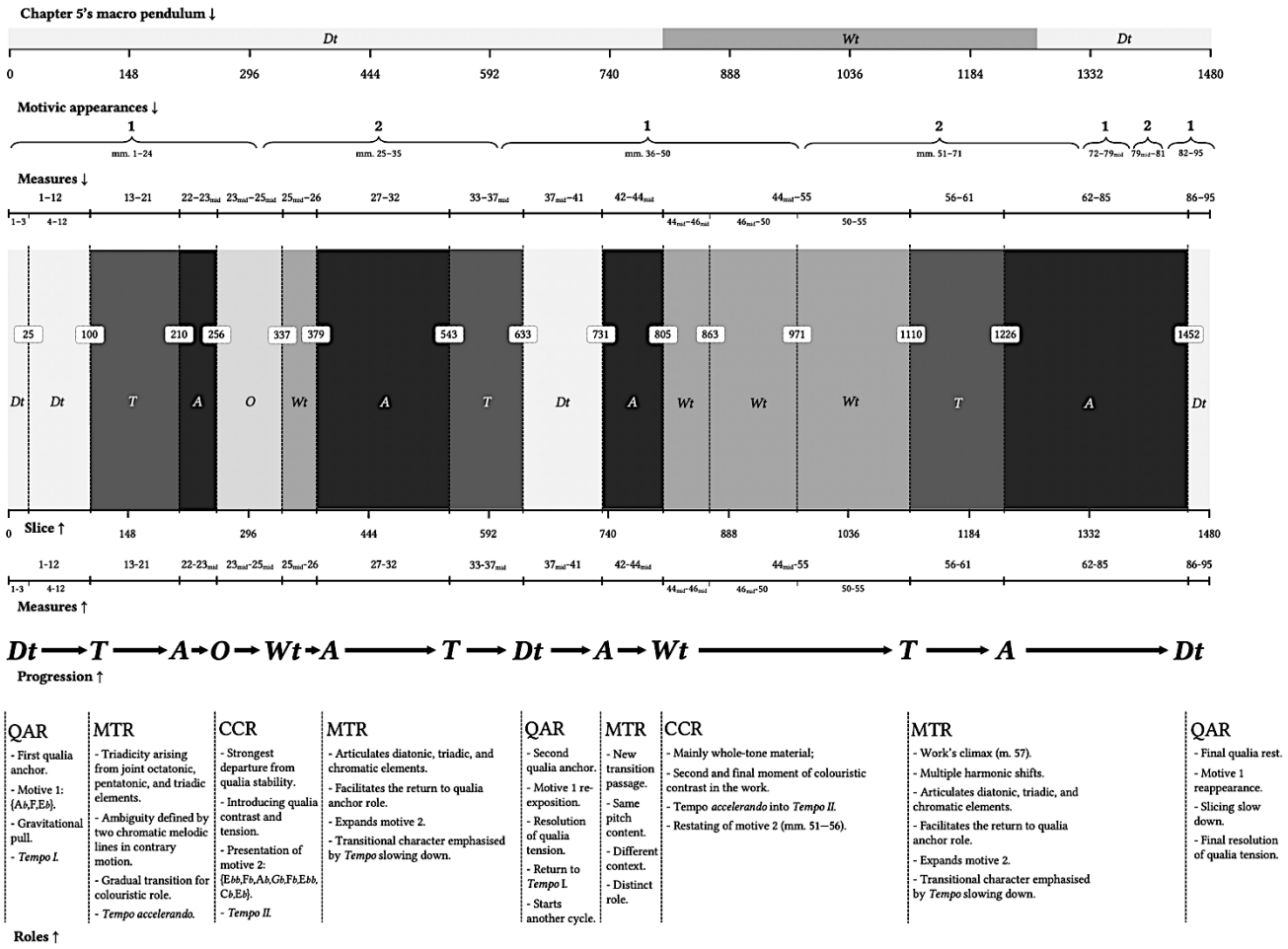


Figure 20. Complete formal analysis of *Reflets dans l'eau* showing the correspondence between macro-level pendular structure, motivic articulations, quality segmentation/progression, and three-role model analysis. Data captured using a peak detection minimum distance of at least 20 pitch-class sets, to better suit the work's rhythmic flow of 15.6 mean pitch-class sets per measure.

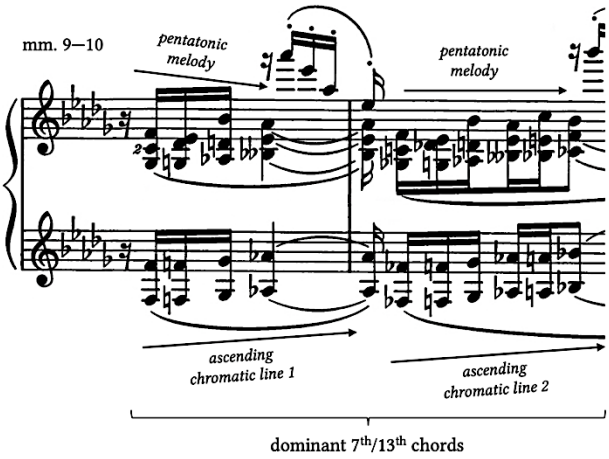


Figure 21. Nested harmonic elements in mm. 9—10 with two ascending chromatic lines supporting dominant seventh/thirteenth progressions, while pentatonic upper-voice melody maintains diatonic orientation.

withholds any return to the diatonic anchor.[16] Having demonstrated the model's coherence across *Mouvement*'s formal design, we turn to *Reflets dans l'eau*, where finer-grained analysis brings to light complexities that aggregate statistics alone cannot capture.

Reflets dans l'eau is built upon the two principal motives shown in Figure 19. Its D♭ major foundation, gradually dissolving into pentatonic cascades, octatonic sequences, and whole-tone passages, epitomises the analytical challenge that harmonic qualities pose to traditional frameworks: listeners readily perceive these qualities and their transformations, yet conventional analysis struggles to represent what constitutes the music's essential character—its quality trajectory. Distance-sensitive segmentation yields a thirteen-stage qualia progression depicted in Figure 20 (see `data/debussy/reflets_segments.csv` in the supplementary repository): $Dt \to T \to A \to O \to Wt \to A \to T \to Dt \to A \to Wt \to T \to A \to Dt$.

Diatonicity surfaces at three structural junctures—opening (mm. 1–12), central return (mm. 37mid–41), and closing (mm. 86–95)—each invariably coinciding with one of the two main melodic motives, notably the three-note figure {A♭,F,E♭} marked with tenuto in mm. 1–2 (Figure 19's upper staff; Howat, 1986, pp. 23–29). Yet this structural overview necessarily conceals finer-grained complexity. The initial diatonic region itself harbours hierarchically nested qualia transformations: bars 9–10 (Figure 21) initiate the first departure from diatonicity through chromatically ascending dominant seventh/thirteenth chord sequences whilst sustaining a pentatonic melodic line. This technique exploits the perceptual disconnect between melodic and harmonic dimensions, preparing the brief yet significant octatonic intrusion (m. 14) that foreshadows the more sustained colouristic section emerging six bars later.

Consider the critical passage spanning mm. 20–26: following the opening diatonic region's transformation through triadicity (mm. 13–21), ambiguity surfaces for a measure and a half before the work's first extended colouristic departure—octatonicity (mm. 23mid–25mid) followed immediately by whole-tone quality (mm. 25mid–26). The pitch content of mm. 13–21 articulates octatonic (m. 14) and pentatonic (mm. 16–17mid)

[16] The fact that this unmediated $Dt \to Wt$ transition accounts for a mere 0.64% of Debussy's overall transitions highlights its exceptional character, confirming the model's predictive power precisely by virtue of how seldom it is violated.

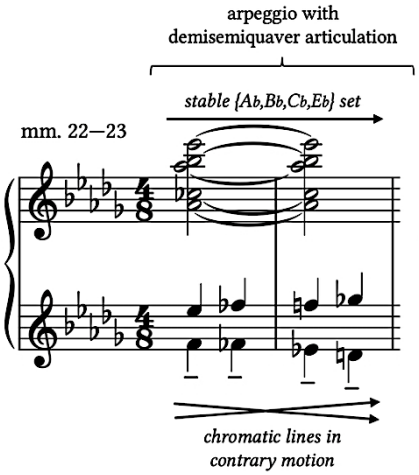


Figure 22. Reduction of mm. 22—23 of *Reflets dans l'eau* highlighting the emergence of qualia ambiguity for the first time in the qualia progression. The musical texture arpeggiates four pitch-class sets comprised of a fixed subset with notes and two chromatic lines moving in contrary motion. These contrapuntal relationships within mm. 22—23 originate the segment's classification as ambiguous.

colours alongside interrupted (m. 18mid, 19mid) and then sequenced (mm. 19–20) triadic constructions such as dominant ninth chords, yielding a globally triadic quality that briefly ushers in the two-measure octatonic segment without first evoking qualia ambiguity in mm. 22–23mid (Figure 22's reduction).[17]

The segment of mm. 23mid–25mid, articulating {D,E♭,F,G♭,A♭,B♭,C♭}, is classified as octatonic on account of its highly symmetrical structure with several tone-semitone steps, furnishing the first colouristic contrast appearance. After these brief two measures, the collection dissolves into a predominantly whole-tone atmosphere ({E𝄫,F♭,G♭,A♭,B♭,C♭}), whose principal function is to introduce the second main melodic motive (Figure 19's lower staff; Howat, 1986, pp. 23–29). This direct *O* → *Wt* progression—bypassing the mediating layer—illustrates how colouristic qualities may chain before re-entering the broader network.

The ensuing ambiguous passage (mm. 27–32) juxtaposes diatonic material with triadic and chromatic elements, operating as a transitional nexus. From this region the music passes through triadicity (mm. 32–37) before achieving diatonic reestablishment (mm. 37–41) with the return of motive 1. A fresh cycle then begins: ambiguity (mm. 41–43) mediates the transition towards the work's most sustained colouristic departure—an eleven-measure whole-tone region (mm. 44–55) constituting nearly twelve per cent of the composition's total duration, corresponding precisely to the *En animant* section.

The diatonic reestablishment and subsequent ambiguous transition exemplify one of the framework's most important capacities: capturing context-dependent qualia roles. Remarkably, mm. 42–43 contain pitch-class material identical to mm. 38–39 (the same vertical sonorities, interval distributions, and slice-by-slice succession) yet receive divergent classifications. Measures 38–39 belong to a segment whose 98 slices across mm. 37–41 exhibit strong perfect-fifth content anchored by C♯–G♯, producing the signature of diatonicity. By contrast, mm. 42–43 belong to a segment whose 74 slices across mm. 41–44 display collapsed perfect-fifth

[17] Note that the triadic and ambiguous segments appear to operate in tandem as transitional devices, progressively introducing deviations from the preceding diatonic content until octatonicity is firmly established. This shift begins with the dominant ninth chords of mm. 18–21, and is subsequently intensified across two measures (mm. 22–23) of qualia ambiguity comprising a fixed set of pitches—{A♭,C♭,B♭,E♭}—and two chromatic, contrary-motion lines—{F↓,F♭↓,E♭↓,D} and {E♭↑,F♭↑, F↑,G♭}—as Figure 22 summarises.

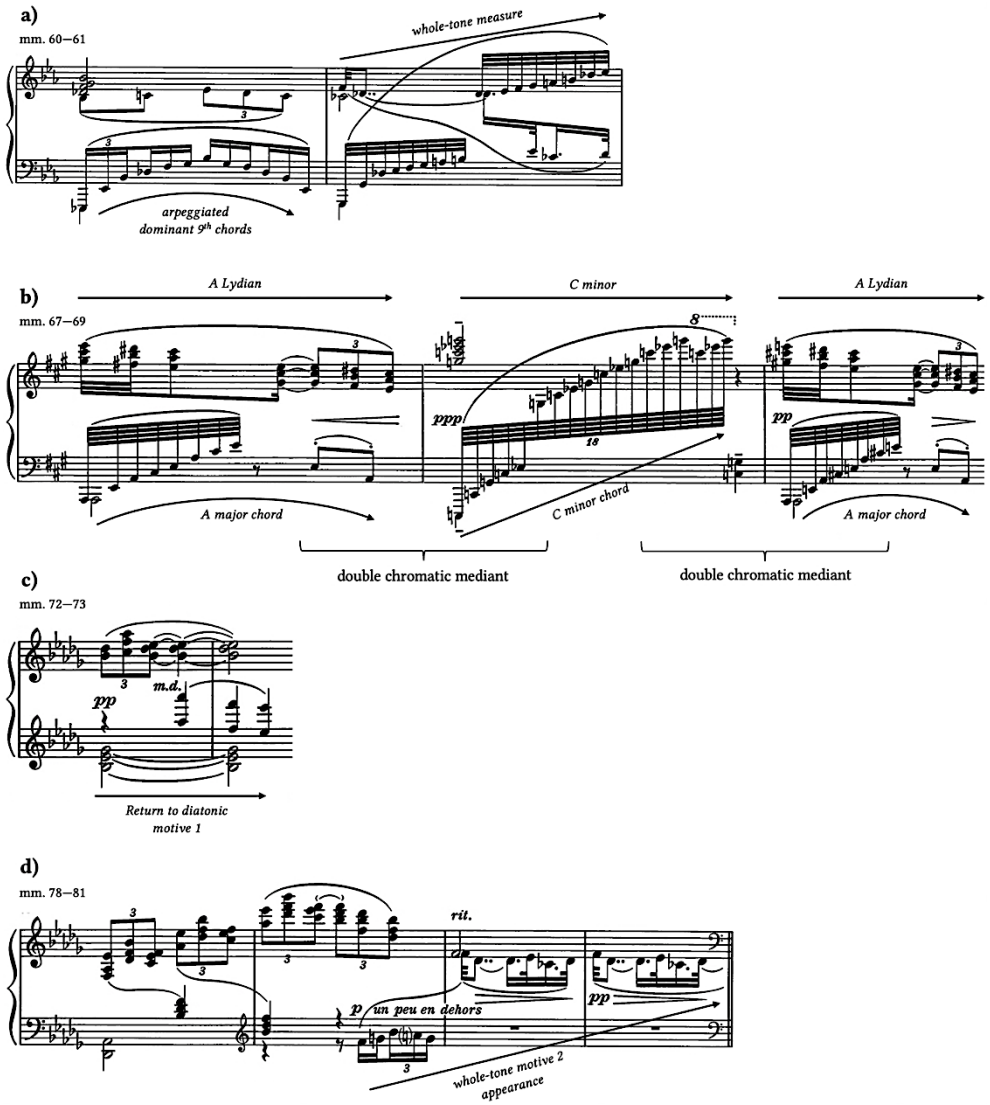


Figure 23. Diverse passages within segment 15's ambiguous region (mm. 62—86) demonstrating continuous qualia fluctuation: a) mm. 60—61, arpeggiated dominant ninths leading into whole-tone; b) mm. 67—69, modal shifts between A Lydian and C minor via double chromatic mediants; c) mm. 72—73, brief diatonic return with motive 1; d) mm. 78—81, whole-tone motive 2 appearance.

content and emergent tritones through incorporation of m. 44's whole-tone material, generating a diffuse character typical of ambiguity. Identical pitch-class material thus serves an anchor function in one context and a mediating function in another—a hallmark of genuinely syntactical discourse, where structural role depends on surrounding context rather than intrinsic content alone.[18]

The work's apex (mm. 57–85) reveals ambiguity's most sophisticated deployment. The appearance of triadicity (mm. 56–61), coinciding with a modulation to E♭ major (Howat, 1986, pp. 23–29), proves short-lived. The subsequent twenty-three measures of ambiguity (mm. 62–85)—the longest single segment in the work—absorb radically heterogeneous material: arpeggiated dominant ninth chords and whole-tone elements (mm. 60, 62; Figure 23a), double chromatic mediants between A Lydian and C minor (mm. 67–69; Figure 23b), the

[18] This observation reinforces a syntactical model grounded in harmonic qualities, since the capacity to identify distinct roles performed by the same structure is a hallmark of inherently syntactical discourses. By way of illustration, the word *book* serves as a direct object in "I read the book" yet functions as the subject in "The book fell".

return of diatonic motive 1 (mm. 72–78; Figure 23c), and a whole-tone restatement of motive 2 (mm. 79mid–81; Figure 23d). That such diverse content coexists under a single ambiguous classification confirms the concept's analytical value: ambiguity is not merely the absence of a predominant quality but an active condition accommodating multiple competing tendencies. The passage ultimately brings the piece full circle towards diatonic stability in its final nine measures.

6. Final remarks and discussion

This study set out to investigate whether composers' deployment of harmonic qualities displays systematic patterns sufficient to constitute a qualia syntax. Having pursued this question through a four-tiered analytical framework—spanning diachronic Zipfian distributions through to work-specific case studies—we now draw together our methodological contributions, evaluate the theoretical implications of our results, and consider the limitations and prospective directions of the proposed model.

6.1 Methodological contributions and general findings

We conducted a systematic analysis of harmonic-quality progressions drawn from composers spanning nearly four centuries within the YCAC corpus. The DFT of pcsets and the FQS were foundational to this endeavour, furnishing the mathematical and geometrical apparatus needed to map not only the six qualities associated with the DFT non-trivial coefficients but also the notion of qualia ambiguity. By treating ambiguity as a distinct analytical category, our approach accommodates musical passages that lack a predominant harmonic quality, thereby extending the framework's applicability to future investigations of diverse post-tonal or atonal repertoires, where such moments may carry structural weight (Mailman, 2015). In addition, our method puts forward an innovative procedure for segmenting musical compositions. Specifically, we devised a distance-sensitive segmentation strategy that pinpoints significant moments of qualia transition via Euclidean distance calculations in the FQS, enabling a perception-aligned mode of structural analysis. This segmentation framework offers a more robust basis for tracing the temporal evolution of harmonic qualities than conventional time-based, window-based, or measure-based approaches.

Our findings reveal evidence of language-like organisation in the harmonic-quality progressions of particular late nineteenth- and early twentieth-century composers—chiefly those associated with musical nationalisms, including Debussy, Ravel, Bartók, and Stravinsky. We also identified a diachronic tendency towards Zipfian distributions, lending support to the idea of a historical evolution from more constrained uses of qualia towards freer, more language-like configurations over successive centuries. In our view, these results carry considerable weight, as they corroborate the hypothesis that a specific syntax governs harmonic qualia, especially in the case of composers whose progressions most closely approximate Zipf's law.

6.2 Theoretical implications: from functional harmony to probabilistic qualities

The three-role model discloses a transformation in the manner through which harmonic elements generate musical syntax. Within tonal music, functional harmony operates through clearly delineated relationships

(tonic, dominant, subdominant) that produce directed motion via tension and resolution. Our analysis points to a broadly analogous system governed by a new organisational logic grounded in harmonic qualities rather than harmonic functions.

Much as tonic furnishes stability in tonal music, diatonicity serves as an anchor within Debussy's system. The relationship, however, differs in a fundamental respect. Tonic functions by virtue of its position within a key-based hierarchy, whereas diatonicity operates as a quality-based reference point realisable at any pitch level. In a parallel vein, while the dominant generates directional tension that demands resolution to the tonic, Debussy's colouristic qualities supply contrast without predetermined resolution paths. The mediating qualities (ambiguity and triadicity) lack any direct counterpart in functional harmony, constituting an entirely new syntactical category that facilitates movement between disparate qualia regions.[19]

Extending the Zipfian evidence set out above, we observe that post-tonal composers such as Debussy did not discard syntax altogether but rather fashioned new structural systems rooted in harmonic qualities instead of functional chord relationships. This shift amounts to a fundamental reconceptualisation, one that finds resonance in Ribeiro-Pereira's (2003, p. 2) theorisation of modulation as an essentially plastic process in which the musical mind shapes harmonic tonal space. Ribeiro-Pereira contends that eighteenth-century theory originally unified what modern discourse has artificially divided: inward harmonic syntax—the manner in which chord progressions articulate a key—and outward key change. His inclusive conception of modulation as harmonic reinterpretation within a single cognitive framework anticipates the type of quality-based transformation our model identifies: just as his notion of plasticity subsumes local chord syntax and global key relations, our analysis discloses how Debussy's diatonicity operates as a quality-based reference point applicable at any pitch level, thereby dissolving the boundary between harmonic function and tonal region. Although we stop short of claiming that the practice our model identifies in Debussy constitutes a complete and formal qualia syntax—a claim that would demand explicit transformation rules exceeding the reach of our statistical evidence—the systematic patterns we detect point to organised structure rather than arbitrary succession. Grasping what kind of structure this is, however, requires attention to its probabilistic character, which raises foundational questions concerning compositional intention versus emergent organisation.

The probabilistic character of our model poses fundamental questions about the relationship between compositional intention and emergent structure. In contrast to the deterministic rules of functional harmony, Debussy's qualia progressions operate through statistical tendencies that accommodate both systematic organisation and creative flexibility.

Two significant caveats warrant emphasis at this juncture. First, throughout this article we use *systematic* to denote the regularity, consistency, and structural coherence evident in the data, without making claims regarding compositional intentionality. The patterns we uncover may reflect conscious design, intuitive

[19] One might tentatively draw a parallel with subdominant function in tonal music where certain harmonies enable smooth transitions between tonal regions without producing the directional tension typical of dominant–tonic progressions. The comparison, however, remains limited: whereas subdominant chords operate within key-based hierarchies and carry predetermined functional meanings, mediating qualities establish neutral territories from which motion to any other quality becomes possible, acting as chameleonic pivots rather than harmonies bearing fixed identities.

choices honed through practice, or emergent properties arising from aesthetic preferences operating beneath explicit awareness—our statistical evidence is unable to adjudicate among these possibilities. The functional roles we propose are therefore analytical frameworks inferred from observed behaviours rather than documented compositional principles. They are nonetheless real, compositionally significant, and analytically productive: systematic organisation need not presuppose systematic planning. Second, our model addresses only harmonic-quality succession, setting aside rhythmic duration, melodic contour, and textural density—properties whose incorporation could refine the transitional probabilities and potentially distinguish between structurally equivalent qualia progressions that diverge in their temporal behaviour.

A more fundamental issue concerns whether the model we advance constitutes a proper qualia syntax. In our judgement, the term *syntax* implies a degree of formality and systematisation that surpasses what our results presently warrant. A complete qualia syntax would necessitate not merely the identification of functional roles but also the codification of generative rules governing qualia progressions. Nonetheless, future inquiries might pursue such rules through grammar induction algorithms capable of learning from qualia progressions and informing a properly syntactical model.

We therefore prefer to characterise our work as a statistically endorsed model of qualia role relationships for Debussy's music—one that fundamentally demonstrates organised structure in the treatment of harmonic qualities and establishes the model as a valuable analytical tool. This probabilistic framework aptly captures the aesthetic aspirations of Impressionism: suggesting rather than stating, implying rather than declaring, achieving coherence through tendency rather than rule.

6.3 Limitations, future directions, and broader significance

Although the model successfully captures patterns within Debussy's harmonic language, its wider applicability and inherent limitations call for careful scrutiny. To begin with, we must acknowledge the technical constraints of our approach. Among the most consequential is the conflation of diatonic and pentatonic moments under *Dt*, which obscures distinctions vital to Debussy's musical language. Pentatonic passages rank among his most characteristic compositional gestures, and their absorption into a broader diatonic category inevitably simplifies the model's representation of his practice. Notwithstanding this limitation, our analysis furnishes valuable empirical support for the three-role model's fundamental premise, confirming how diatonicity anchors the structure whilst ambiguity mediates transitions to the colouristic passages that produce Debussy's distinctive floating and impressionistic sound.

Turning to future prospects, subsequent research might profitably extend this probabilistic framework by quantifying the conditions under which harmonic qualities assume non-typical functions. The direct $Dt \rightarrow O$ connection identified in *Reflets dans l'eau*—bypassing the mediating layer predicted by the model—exemplifies such exceptions: clarifying when and why colouristic qualities connect directly to the anchor, rather than via ambiguity or triadicity, would sharpen the model's account of Debussy's compositional logic. Comparative analyses of other composers from the same period would be especially worthwhile, in order to ascertain whether this three-role framework captures a broader trend in early twentieth-century harmonic practice or

remains specific to Debussy's individual style. Preliminary observations suggest that Ravel's harmonic organisation, while sharing the tripartite structure, exhibits different internal relationships.

The identification of probabilistic patterns in Debussy's harmonic language ultimately sheds light on his singular position at the crossroads of tradition and innovation. By delineating general tendencies whilst preserving contextual flexibility, the model reveals how Debussy achieved a mode of composition that was at once structured and experimental. The anchor, transitional, and colouristic roles supply a theoretical framework that captures both the structural coherence and the expressive breadth characteristic of early twentieth-century harmonic innovation.

At a broader level, this analysis illustrates how computational methods can uncover syntactical structures in post-tonal music that might otherwise remain concealed. By treating harmonic qualities as functional units amenable to statistical analysis, we bring to light organisational principles that transcend traditional theoretical categories. This equilibrium between order and freedom continues to define Debussy's lasting contribution to Western musical language and provides fertile terrain for further analytical inquiry. As music theory continues to engage with the diverse languages of twentieth- and twenty-first-century composition, probabilistic models may prove increasingly indispensable for understanding how composers forge coherence without adhering to predetermined rules.

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Supplemental material

Supplemental material, including source code, transition count data, dendrograms, and analytical outputs, is available at: https://github.com/sml-pereira/qualia-syntax or https://doi.org/10.5281/zenodo.19429590.

References

Amiot, E. (2016). *Music Through Fourier Space*. Springer International Publishing. https://doi.org/10.1007/978-3-319-45581-5

Amiot, E. (2017). Interval Content vs. DFT. In O. A. Agustín-Aquino, E. Lluis-Puebla, & M. Montiel (Eds), *Mathematics and Computation in Music* (Vol. 10527, pp. 151–166). Springer International Publishing. https://doi.org/10.1007/978-3-319-71827-9_12

Anderson, C. J., & Schutz, M. (2023). Understanding Feature Importance in Musical Works: Unpacking Predictive Contributions to Cluster Analyses. *Music & Science*, *6*, 20592043231216257. https://doi.org/10.1177/20592043231216257

Bernardes, G., Carvalho, N., & Pereira, S. (2023). FluidHarmony: Defining an equal-tempered and hierarchical harmonic lexicon in the Fourier space. *Journal of New Music Research*, *51*(2–3), 142–161. https://doi.org/10.1080/09298215.2023.2202641

Bernardes, G., Cocharro, D., Caetano, M., Guedes, C., & Davies, M. E. P. (2016). A multi-level tonal interval space for modelling pitch relatedness and musical consonance. *Journal of New Music Research*, *45*(4), 281–294. https://doi.org/10.1080/09298215.2016.1182192

Boss, J. F. (2019). *Schoenberg's atonal music: Musical idea, basic image and specters of tonal function*. Cambridge, United Kingdom.

Caplin, W. E. (2001). *Classical form: A theory of formal functions for the instrumental music of Haydn, Mozart, and Beethoven*. Oxford Univ. Press.

Chiu, M. (2021). Macroharmonic Progressions through the Discrete Fourier Transform: An Analysis from Maurice Duruflé's Requiem. *Music Theory Online*, *27*(3). https://doi.org/10.30535/mto.27.3.1

Chomsky, N. (2015). *Aspects of the theory of syntax* (50th Anniversary Edition). The MIT Press.

Conklin, D., & Anagnostopoulou, C. (2001). Representation and Discovery of Multiple Viewpoint Patterns. *Proceedings of the 2001 International Computer Music Conference*, 479–485. http://hdl.handle.net/2027/spo.bbp2372.2001.092

Conklin, D., & Witten, I. H. (1995). Multiple viewpoint systems for music prediction. *Journal of New Music Research*, *24*(1), 51–73. https://doi.org/10.1080/09298219508570672

Dunsby, J. (Ed.). (1993). *Early twentieth century music*. Blackwell.

Forte, A. (1980). *The structure of atonal music* (Nachdr.). Yale Univ. Press.

Forte, A., & Gilbert, S. E. (1982). *Introduction to Schenkerian analysis*. Norton.

Harasim, D., Affatato, G., & Moss, F. C. (2022). midiVERTO: A Web Application to Visualize Tonality in Real Time. In M. Montiel, O. A. Agustín-Aquino, F. Gómez, J. Kastine, E. Lluis-Puebla, & B. Milam (Eds), *Mathematics and Computation in Music* (Vol. 13267, pp. 363–368). Springer International Publishing. https://doi.org/10.1007/978-3-031-07015-0_31

Harding, J. D. (2021). *Applications of the Discrete Fourier Transform to Music Analysis* [PhD]. Florida State University.

Hasty, C. F. (1984). Phrase Formation in Post-Tonal Music. *Journal of Music Theory*, *28*(2), 167. https://doi.org/10.2307/843531

Hook, J. (2023). *Exploring Musical Spaces: A Synthesis of Mathematical Approaches* (1st edn). Oxford University Press. https://doi.org/10.1093/oso/9780190246013.001.0001

Howat, R. (1986). *Debussy in Proportion: A Musical Analysis*. Cambridge University Press. https://www.cambridge.org/us/universitypress/subjects/music/twentieth-century-and-contemporary-music/debussy-proportion-musical-analysis?format=PB&isbn=9780521311458

Huron, D. B. (2006). *Sweet Anticipation: Music and the Psychology of Expectation*. MIT Press. https://mitpress.mit.edu/9780262582780/sweet-anticipation/

Inselberg, A., & Dimsdale, B. (1987). Parallel Coordinates for Visualizing Multi-Dimensional Geometry. In T. L. Kunii (Ed.), *Computer Graphics 1987* (pp. 25–44). Springer Japan. https://doi.org/10.1007/978-4-431-68057-4_3

Jackendoff, R. (2004). *Foundations of language: Brain, meaning, grammar, evolution* (1. publ. in paperback, 2. impr). Oxford Univ. Press.

Johnson, S. C. (1967). Hierarchical clustering schemes. *Psychometrika*, *32*(3), 241–254. https://doi.org/10.1007/BF02289588

Kramer, J. D. (1988). *The Time of Music: New Meanings, New Temporalities, New Listening Strategies*. Schirmer Books ; Collier Macmillan Publishers.

Krumhansl, C. L. (2005). The geometry of musical structure: A brief introduction and history. *Computers in Entertainment*, *3*(4), 1–14. https://doi.org/10.1145/1095534.1095542

Lerdahl, F. (2001). *Tonal pitch space*. Oxford University Press.

Lerdahl, F. (2013). Musical Syntax and Its Relation to Linguistic Syntax. In M. A. Arbib (Ed.), *Language, Music, and the Brain* (pp. 257–272). The MIT Press. https://doi.org/10.7551/mitpress/9780262018104.003.0010

Lerdahl, F., & Jackendoff, R. (1990). *A generative theory of tonal music* (4. print). MIT Press.

Lesure, F. (1977). *Catalogue De L'œuvre De Claude Debussy*. Minkoff [diffusion Frankelve].

Lewin, D. (1959). Re: Intervallic Relations between Two Collections of Notes. *Journal of Music Theory*, *3*(2), 298. https://doi.org/10.2307/842856

Lewin, D. (2001). Special Cases of the Interval Function between Pitch-Class Sets X and Y. *Journal of Music Theory*, *45*(1), 1–29. https://doi.org/10.2307/3090647

Mailman, J. B. (2015). Schoenberg's Chordal Experimentalism Revealed through Representational Hierarchy Association (RHA), Contour Motives, and Binary State Switching. *Music Theory Spectrum*, *37*(2), 224–252. https://doi.org/10.1093/mts/mtv015

Manaris, B., Romero, J., Machado, P., Krehbiel, D., Hirzel, T., Pharr, W., & Davis, R. B. (2005). Zipf's Law, Music Classification, and Aesthetics. *Computer Music Journal*, *29*(1), 55–69. https://doi.org/10.1162/comj.2005.29.1.55

Mandelbrot, B. (1953). An informational theory of the structure of language based upon the theory of the statistical matching of messages and coding. In W. Jackson (Ed.), *Proceedings of the Symposium on Applications of Communication Theory* (pp. 486–582).

Martins, J. O. (2015). Bartók's Polymodality. *Journal of Music Theory*, *59*(2), 273–320. https://doi.org/10.1215/00222909-3136020

Martins, J. O. (2023). Lutosławski's Mid-Century Harmony. *Journal of Music Theory*, *67*(2), 209–249. https://doi.org/10.1215/00222909-10699703

Morgan, R. P. (1991). *Anthology of twentieth-century music*. W.W. Norton.

Moss, F. C., Neuwirth, M., Harasim, D., & Rohrmeier, M. (2019). Statistical Characteristics of Tonal Harmony: A Corpus Study of Beethoven's String Quartets. *PLOS ONE*, *14*(6), e0217242. https://doi.org/10.1371/journal.pone.0217242

Pearce, M., & Rohrmeier, M. (2018). Musical Syntax II: Empirical Perspectives. In R. Bader (Ed.), *Springer Handbook of Systematic Musicology* (pp. 487–505). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-55004-5_26

Pereira, S., Affatato, G., Bernardes, G., & Moss, F. C. (2024). Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity. In T. Noll, M. Montiel, F. Gómez, O. C. Hamido, J. L. Besada, & J. O. Martins (Eds.), *Mathematics and Computation in Music* (Vol. 14639, pp. 317–329). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-60638-0_25

Pereira, S., Bernardes, G., & Martins, J. O. (2025). Qualia Motion in Fourier Space: Formalizing Linear, Nondirected, and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1. *Music Theory Spectrum*. https://doi.org/10.1093/mts/mtaf016

Quinn, I. (1999). The combinatorial model of pitch contour. *Music Perception*, *16*(4), 439–456. https://doi.org/10.2307/40285803

Quinn, I. (2006). General Equal-Tempered Harmony (Introduction and Part I). *Perspectives of New Music*, *44*(2), 114–158.

Quinn, I. (2007). General Equal-Tempered Harmony: Parts 2 and 3. *Perspectives of New Music*, *45*(1), 4–63.

Rameau, J.-P. (1971). *Treatise on harmony* (P. Gossett, Trans.). Dover Publ.

Ribeiro-Pereira, J. M. (2003). *A Theory of Harmonic Modulation: The Plastic Model of Tonal Syntax and the Major-Minor Key System* (1st edn). Instituto Politécnio do Porto. https://edicoes.ipp.pt/index.php/books/catalog/book/11

Riemann, H. (1896). *Harmony simplified: Or, the theory of the tonal functions of chords* (Augener's edition no. 9198). Augener.

Rohrmeier, M. (2011). Towards a generative syntax of tonal harmony. *Journal of Mathematics and Music*, *5*(1), 35–53. https://doi.org/10.1080/17459737.2011.573676

Rohrmeier, M. A., & Moss, F. C. (2021). A Formal Model of Extended Tonal Harmony. *Proceedings of the 22nd International Society for Music Information Retrieval Conference*, 569–578. https://doi.org/10.5281/ZENODO.5624635

Rohrmeier, M., & Cross, I. (2008). Statistical properties of tonal harmony in Bach's chorales. *Proceedings of the 10th International Conference on Music Perception and Cognition*, 619–627.

Rohrmeier, M., & Graepel, T. (2012). Comparing feature-based models of harmony. *Proceedings of the 9th International Symposium on Computer Music Modeling and Retrieval*, 357–370.

Rohrmeier, M., & Pearce, M. (2018). Musical Syntax I: Theoretical Perspectives. In R. Bader (Ed.), *Springer Handbook of Systematic Musicology* (pp. 473–486). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-55004-5_25

Schenker, H., & Oster, E. (with American Council of Learned Societies). (2001). *Free composition: Volume III of New musical theories and fantasies = Der freie Satz*. Pendragon Press.

Somer, A. (2005). Musical Syntax in the Sonatas of Debussy: Phrase Structure and Formal Function. *Music Theory Spectrum*, *27*(1), 67–96. https://doi.org/10.1525/mts.2005.27.1.67

Temperley, D. (2007a). *Music and probability*. MIT Press.

Temperley, D. (2007b). The Tonal Properties of Pitch-Class Sets: Tonal Implication, Tonal Ambiguity, and Tonalness. In *Tonal Theory for the Digital Age* (Vol. 15, pp. 24–38). Center for Computer Assisted Research in the Humanities at Stanford University. https://www.ccarh.org/publications/books/cm/vol/15/cm15-02-temperley.pdf

Tymoczko, D. (2011). *A Geometry of Music: Harmony and Counterpoint in the Extended Common Practice*. Oxford University Press. https://global.oup.com/academic/product/a-geometry-of-music-9780195336672?q=978-0-19-533667-2&cc=pt&lang=en

Viaccoz, C., Harasim, D., Moss, F. C., & Rohrmeier, M. (2023). Wavescapes: A Visual Hierarchical Analysis of Tonality Using the Discrete Fourier Transform. *Musicae Scientiae*, *27*(2), 390–427. https://doi.org/10.1177/10298649211034906

Voss, R. F., & Clarke, J. (1975). '1/fnoise' in Music and Speech. *Nature*, *258*(5533), 317–318. https://doi.org/10.1038/258317a0

Voss, R. F., & Clarke, J. (1978). ''1/ f Noise'' in Music: Music from 1/f Noise. *The Journal of the Acoustical Society of America*, *63*(1), 258–263. https://doi.org/10.1121/1.381721

Ward, J. H. (1963). Hierarchical Grouping to Optimize an Objective Function. *Journal of the American Statistical Association*, *58*(301), 236–244. https://doi.org/10.1080/01621459.1963.10500845

White, C. W., & Quinn, I. (2016). The Yale-Classical Archives Corpus. *Empirical Musicology Review*, *11*(1), 50–58. https://doi.org/10.18061/emr.v11i1.4958

Yust, J. (2015a). Applications of DFT to the Theory of Twentieth-Century Harmony. In T. Collins, D. Meredith, & A. Volk (Eds.), *Mathematics and Computation in Music* (Vol. 9110, pp. 207–218). Springer International Publishing. https://doi.org/10.1007/978-3-319-20603-5_22

Yust, J. (2015b). Schubert's Harmonic Language and Fourier Phase Space. *Journal of Music Theory*, *59*(1), 121–181. https://doi.org/10.1215/00222909-2863409

Yust, J. (2016). Special Collections. *Journal of Music Theory*, *60*(2), 213–262. https://doi.org/10.1215/00222909-3651886

Yust, J. (2019). Stylistic information in pitch-class distributions. *Journal of New Music Research*, *48*(3), 217–231. https://doi.org/10.1080/09298215.2019.1606833

Yust, J. (2020). Dimensions of Atonality: A Response and Extension of von Hippel and Huron (2020). *Empirical Musicology Review*, *15*(1–2), 119–127. https://doi.org/10.18061/emr.v15i1-2.7832

Yust, J., Lee, J., & Pinsky, E. (2022). A Clustering-Based Approach to Automatic Harmonic Analysis: An Exploratory Study of Harmony and Form in Mozart's Piano Sonatas. *Transactions of the International Society for Music Information Retrieval*, *5*(1), 113–128. https://doi.org/10.5334/tismir.114

Zanette, D. H. (2004). *Zipf's law and the creation of musical context* (No. arXiv:cs/0406015). arXiv. https://doi.org/10.48550/arXiv.cs/0406015

Zipf, G. K. (2012). *Human behavior and the principle of least effort: An introduction to human ecology*. Martino Publishing [u.a.].